AI-POWERED EMPLOYABILITY ASSESSMENT AND CAREER READINESS PLATFORM USING RESUME ANALYSIS AND SKILL GAP ASSESSMENT

Project-I Report

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

P.S.R. ENGINEERING COLLEGE, SIVAKASI

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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OCTOBER 2026

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ACKNOWLEDGEMENT

We take this opportunity to thank all those who helped towards the successful completion of this project. At the very outset, we thank the Almighty for His profuse blessings showered on us. We thank our beloved parents whose encouragement and support helped us to complete our project successfully.

It is our greatest pleasure to express our sincere thanks to **Thiru R. SOLAISAMY,** Correspondent and Director, **Er. S. VIGNESWARI ARUNKUMAR B.Tech.,** P.S.R. Engineering College, Sivakasi, for providing the required facilities and suitable infrastructure to carry out this project.

It is our privilege to convey our sincere thanks to our respected and honourable Principal **Dr. J. S. SENTHIL KUMAAR, M.E., Ph.D.,** for giving us permission to do this project in our institution.

We wish to extend and express our sincere thanks to our adored Head of the Department, **Dr. C. NATARAJAN, M.E., Ph.D., Associate Professor** for his excellent guidance and motivation to complete our project proficiently.

We also wish to express our hearty thanks to our Project Guide, **Mrs. V. DEVI NANDHINI, M.E., Assistant Professor** for his excellent guidance and constant encouragement during this project work.

We also wish to express our sincere thanks to our Esteemed Project Coordinators **Dr. C. NATARAJAN, M.E., Ph.D., Associate Professor And Miss. G. SEENIAMMAL, M.E., Assistant Professor** for their continuous guidance and valuable suggestions, without whom this project would not have become a reality.

We are also bound to thank all the Teaching and Non-Teaching staff members of our **Artificial Intelligence and Data Science Department** who supported us with their wishes and prayers that brought success and completion to our project.

ABSTRACT

Career Compass AI is an AI-powered employability assessment and career readiness platform designed to help students evaluate their resumes, identify skill gaps and improve their preparation for targeted job roles. The system analyses uploaded resumes to assess resume quality, ATS-oriented scores, technical skills and profile strengths. It processes selected job descriptions to extract relevant requirements and compares them with the candidate's profile to identify matching and missing skills. Based on the identified skill gaps, the platform recommends suitable projects and relevant job roles to support practical skill development. A role-specific mock assessment evaluates the student's technical knowledge and generates performance scores for further analysis. The system tracks assessment performance across multiple attempts and provides skill-wise analytics to identify strengths and areas requiring improvement. A personalized career roadmap guides students through the skills and learning activities that need further attention. Finally, a career-readiness indicator combines profile alignment, skill coverage and assessment performance to provide an overview of the student's preparation status. The complete platform integrates these modules through a web-based interface developed using React and TypeScript, with a FastAPI backend and database support. The proposed framework establishes a continuous career-preparation process through resume analysis, skill-gap identification, practical learning, assessment and progress monitoring, helping students make informed decisions about their employability development.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	ARTIFICIAL INTELLIGENCE
ML	MACHINE LEARNING
ATS	APPLICANT TRACKING SYSTEM
JD	JOB DESCRIPTION
LLM	LARGE LANGUAGE MODEL
API	APPLICATION PROGRAMMING INTERFACE
UI	USER EXPERIENCE
UX	REPRESENTATIONAL STATE TRANSFER
PDF	PORTABLE DOCUMENT FORMAT
OCR	OPTICAL CHARACTER RECOGNITION
SQL	STRUCTURED QUERY LANGUAGE
DB	DATABASE
HTTP	HYPERTEXT TRANSFER PROTOCOL
REST	REPRESENTATIONAL STATE TRANSFER
JSON	JAVASCRIPT OBJECT NOTATION

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The rapid growth of digital recruitment and the increasing demand for specialized technical skills have created a need for effective methods to understand the relationship between a candidate's profile and the requirements of a particular job role. A resume generally contains information about a student's education, technical skills, projects, certifications, experience and achievements, but this information is usually presented as unstructured text. Natural Language Processing (NLP) techniques can be used to extract meaningful information from resumes and identify relevant skills and qualifications [1], [3]. Students also need to understand which skills they are missing, which skills should be prioritized, what practical projects can help them develop those skills, and whether their knowledge improves after learning. Career Compass AI is therefore designed as a student-oriented employability assessment and career-readiness platform that connects resume analysis, job-description analysis, skill-gap identification, project recommendation, job-role recommendation, mock assessment, skill analytics and personalized roadmap generation into a continuous workflow. The system follows the process of analysing the student's current profile, identifying the difference between existing and required skills, providing practical improvement recommendations, evaluating the student's progress and updating the learning direction based on assessment performance. In this way, the proposed platform moves beyond a static resume-analysis approach and provides a structured process for students to understand their current employability profile and prepare for target job roles.

1.2 BACKGROUND

The transition from academic education to employment requires students to understand the skills expected by organizations and compare them with their existing knowledge and experience. Job descriptions provide important information about technical skills, frameworks, tools, databases, cloud technologies, educational qualifications and experience requirements, while resumes represent the skills and experiences possessed by candidates. Research has explored the extraction of skills from job postings and the use of NLP techniques for understanding labour-market requirements [7], while several studies have investigated automated resume analysis and candidate–job matching [1], [11], [14]. Recent approaches have also considered

semantic relationships between resumes and job descriptions rather than relying only on exact keyword matching [3], [9]. At the same time, skill recommendation and personalized learning systems have attempted to connect identified gaps with suitable learning opportunities [5], [16]. These developments provide the technical foundation for Career Compass AI, where resume information, job requirements and student assessment performance are considered together to support employability preparation.

Recent research has further examined intelligent resume analysis and skill matching using NLP-based techniques to improve the identification of relevant candidate competencies [5]. Resume recommendation approaches have also explored automated feedback mechanisms to help candidates improve the quality and relevance of their profiles [2]. Other studies have investigated semantic relationships between candidate profiles and job requirements to achieve more meaningful matching beyond simple keyword overlap [7]. Research on resume semantics has also connected job and course recommendations with candidate profiles, supporting more personalized career guidance [6]. Student placement prediction and skill recommendation approaches have highlighted the importance of identifying relevant competencies for improving employability outcomes [8]. Hypergraph-based resume–job matching has further explored relationships among multiple skills and job requirements for improved matching representation [7]. These studies indicate that employability systems can benefit from combining profile analysis with structured skill relationships and recommendations. However, many existing approaches primarily focus on one or two stages such as resume screening, matching or recommendation rather than providing a continuous preparation workflow. Career Compass AI addresses this gap by connecting resume analysis, job matching, skill-gap identification, project recommendations, role-specific assessment and progress tracking within a unified platform.

1.3 PROBLEM STATEMENT

Students often find it difficult to determine whether their current resume and technical skills are suitable for a particular job role. A student may have completed courses, projects and certifications but may not know which of these skills are relevant to a specific job description or which important skills are still missing. Existing resume screening and job-matching approaches mainly focus on extracting information, calculating similarity or ranking candidates [1], [6], [12]. Although these approaches are useful for recruitment-oriented analysis, they do not always provide a complete improvement path for students. Skill recommendation systems can identify additional

skills [5], while job and course recommendation frameworks can provide learning-oriented suggestions [10], [16], but the different activities are often handled separately. Therefore, Career Compass AI addresses the problem of integrating resume analysis, job requirement analysis, skill-gap identification, practical project recommendation, assessment and progress tracking into a unified student-focused system.

Recent resume analysis approaches have demonstrated the usefulness of NLP techniques for extracting candidate skills and comparing them with job requirements [5]. Semantic resume–job matching methods have further shown that relationships between related skills can provide better alignment than direct keyword comparison alone [7]. Research on resume semantics has also explored combining candidate information with job and course recommendations to support personalized career decisions [6]. Skill recommendation studies indicate that identifying missing competencies can help candidates understand the areas that require further development [5]. Student placement and skill recommendation approaches have emphasized the importance of aligning technical competencies with the requirements of targeted employment opportunities [8]. Other research has explored structured relationships between resumes and job requirements to improve the representation of candidate–job compatibility [7]. These studies demonstrate that automated analysis can provide useful information for understanding employability gaps. However, identifying a gap alone may not be sufficient for students unless the system also provides practical activities and measurable assessment. Career Compass AI therefore extends the analysis process by connecting identified skill gaps with project recommendations and role-specific mock assessments. The assessment results can then be used to support skill analytics and personalized roadmap generation, creating a continuous preparation cycle for students.

1.4 MOTIVATION

The motivation for developing Career Compass AI is to provide students with a clearer understanding of their career preparation status and the actions required to improve it. A simple resume score or job-match percentage does not explain what a student should do after receiving the result. Research on learning analytics has shown that learner performance data can be used to understand learning progress and support appropriate interventions [17], while personalized learning research highlights the importance of adapting learning requirements [20], [21]. Similarly, adaptive assessment approaches can use performance information to influence subsequent learning recommendations [18].

Based on these ideas, recommending practical projects, conducting role-specific assessments and tracking performance over multiple attempts. The motivation is therefore to transform a one-time resume evaluation into a continuous process of analysis, practice, assessment and improvement.

1.5 OBJECTIVES

The primary objective of Career Compass AI is to develop an integrated platform that helps students analyse their employability profile and prepare for specific job roles. The system aims to analyse resumes and extract relevant information such as skills, education, projects and experience; analyse job descriptions to identify required and preferred skills; compare resume skills with job requirements; identify and prioritize skill gaps; recommend practical projects that address multiple missing skills; suggest suitable job roles based on the student's profile; conduct role-specific mock assessments; calculate skill-wise performance; track improvement across assessment attempts; and generate a personalized roadmap based on identified gaps and assessment results. These objectives combine concepts explored in intelligent resume recommendation [4], skill recommendation [5], job-market skill extraction [7], adaptive assessment [18] and learning analytics [17] to create a single career preparation workflow.

The system is motivated by the need to connect assessment results with specific skill-development activities. Performance trends can help students recognize whether their preparation is improving over time [17]. Personalized recommendations can further direct students towards skills that require greater attention [20]. Adaptive assessment concepts support the use of previous performance when planning subsequent learning activities [18]. Project-based recommendations can provide students with opportunities to apply newly developed technical skills in practical situations. Repeated assessments can also help students evaluate the effectiveness of their preparation. The integration of these activities reduces the gap between identifying a weakness and taking corrective action. Therefore, the proposed system aims to provide a structured cycle of analysis, practice, assessment and improvement rather than a one-time evaluation.

1.6 SCOPE

The scope of Career Compass AI covers the major stages involved in student-oriented career preparation, beginning with resume submission and extending to skill analysis, assessment and roadmap generation. The platform is intended to process resume information, analyse selected job requirements, identify matching and missing skills, recommend relevant projects and job roles, conduct mock assessments and visualize skill progress. The system can also generate a learning roadmap based on the identified skill priorities and assessment performance.

Personalized learning research indicates that learning paths can be adapted according to learner characteristics, goals and competency levels [21], while adaptive learning systems can use real-time performance information to improve recommendations [22]. Career Compass AI applies these principles within an employability-focused workflow. The system is primarily intended for college students, final-year students and fresh graduates preparing for technical job opportunities. The career-readiness value produced by the system represents profile alignment and preparation status and should not be interpreted as a guaranteed prediction of employment or job selection.

The scope also includes maintaining assessment history so that students can review their performance across multiple attempts. Skill-wise analytics can help identify areas where improvement is required and support more focused preparation [17]. The project recommendation component provides practical opportunities for students to apply skills identified as missing from their profiles. Job-role recommendations further help students understand the roles that are more closely aligned with their current competencies.

The roadmap component organizes improvement areas into a structured preparation sequence based on identified priorities. These features allow the system to connect profile analysis with practical skill development rather than providing only static evaluation results. The platform is designed primarily as a student-support system for career preparation and self-assessment. Its outputs are intended to provide guidance and awareness rather than replace professional recruitment decisions. Overall, the defined scope establishes a continuous workflow from resume analysis to skill development, assessment and career-readiness evaluation.

1.7 APPLICATIONS

Career Compass AI can be applied in college placement preparation, career guidance centres, student skill-development programmes and individual job-preparation activities. Students can use the platform to compare their current skills with the requirements of a target role, identify important gaps, select practical projects for improvement and evaluate their knowledge through mock assessments. Educational institutions can also use such a system to understand common skill deficiencies among students and support targeted training activities.

appropriate interventions [17], while AI-based personalized learning systems provide opportunities for individualized recommendations and assessment [20]. The proposed platform can therefore act as a bridge between academic learning and employment preparation by connecting resume information, job requirements, practical skill development, assessment performance and personalized learning guidance within one system.

The platform can also support students in preparing for different technical roles by providing role-specific analysis and assessment. The identified skill gaps can be used to guide students towards relevant projects and learning activities. Assessment history allows students to monitor their progress and understand changes in their performance over time [17].

Placement cells can potentially use aggregated insights to identify frequently occurring skill gaps and plan focused training programmes. Career guidance centres can use the platform as a supporting tool for individual counselling and preparation. By combining these activities, Career Compass AI provides a structured approach for continuous employability development.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Recent developments in Artificial Intelligence and Natural Language Processing have increased the use of automated techniques for analysing resumes, job descriptions, skills and learning requirements. Research in this area has focused on extracting meaningful information from resumes, identifying skills from job descriptions, matching candidates with suitable job roles and recommending additional skills or learning resources [1], [4], [5]. Resume screening systems have explored NLP-based feature extraction and matching methods, while more recent approaches have introduced semantic representations and machine learning techniques to improve the understanding of candidate profiles [9], [11]. Job-market research has also examined the extraction and classification of skills from large collections of job postings to understand changing employer requirements [7]. In parallel, learning analytics and personalized learning research have investigated how learner performance can be used to provide individualized learning recommendations and feedback [17], [20], [21]. These research areas provide the foundation for Career Compass AI, which combines resume analysis, job-description analysis, skill-gap identification, recommendation, assessment and progress tracking within a single employability-oriented workflow.

Further, NLP-based approaches support the extraction of technical and non-technical information from unstructured candidate profiles. Semantic matching methods can help identify relationships between skills even when different terms are used to describe similar competencies. Recommendation-based systems can use the identified skill gaps to suggest suitable career paths, projects and learning activities [2], [3], [6]. Research on intelligent resume analysis has also demonstrated the importance of combining resume information with job requirements rather than evaluating resumes independently [5], [8]. Skill recommendation studies further highlight the need to connect identified competencies with appropriate development opportunities [3], [6]. At the same time, learning analytics can provide useful information about learner progress by analysing assessment performance over multiple attempts [17]. Personalized learning approaches can use such information to adapt learning activities according to individual requirements [20], [21].

These concepts support the integration of assessment and roadmap generation into the proposed platform. By connecting profile analysis with practical skill development and continuous assessment, Career Compass AI extends the functionality of conventional resume-analysis systems. The overall approach therefore brings together existing research areas into a unified workflow designed specifically for student employability preparation.

2.2 RESUME ANALYSIS USING NLP

Resume analysis is an important research area in automated recruitment because resumes contain information in different formats and structures, making manual processing time-consuming when large numbers of candidates are involved. NLP techniques can be applied to identify sections such as education, experience, technical skills, projects and certifications from resume documents. Earlier systems primarily relied on keyword extraction, rule-based processing and feature matching, while later approaches introduced machine learning, contextual representations and transformer-based methods to improve information extraction and candidate profiling [1], [2], [4]. Studies have also investigated semantic resume representations to identify relationships between candidate competencies and job requirements [3], [11]. These approaches demonstrate that resume processing involves more than simply reading individual keywords because the same skill may be expressed using different terms or abbreviations. For example, technologies such as ReactJS and React may represent the same underlying skill, requiring normalization before comparison. Career Compass AI follows this general direction by extracting resume text, identifying relevant skills and organizing the information into structured data that can subsequently be used for job comparison and skill-gap analysis.

Research has also focused on improving the accuracy of skill extraction from resumes by using predefined skill taxonomies and semantic similarity techniques [5], [9]. Such methods can reduce the limitations of exact keyword matching when candidates use alternative names for the same technology. Resume analysis can also provide a structured representation of candidate capabilities that can be reused across different stages of the recruitment or career-guidance process. Studies on intelligent resume recommendation have shown that extracted candidate information can support suitable job and skill recommendations [2], [6]. In addition, semantic approaches can improve the comparison between candidate profiles and job descriptions by considering the contextual relationship between terms [11].

This is particularly useful when a job description contains skills that are represented differently in a candidate's resume. The extracted resume information can therefore serve as the foundation for identifying both existing competencies and potential skill gaps. In Career Compass AI, the structured resume information is further used for role-specific matching, project recommendation and assessment generation. This allows the resume analysis component to contribute directly to the later stages of career preparation. Thus, resume processing forms the initial and essential stage of the proposed employability assessment workflow.

2.3 JOB DESCRIPTION ANALYSIS

Job descriptions provide information about the skills, technologies, qualifications and experience expected for a particular role, and analysing these requirements is necessary for meaningful candidate–job comparison. Research on job-market analysis has examined methods for extracting and classifying skills from job postings and identifying trends in employer requirements [7]. Traditional approaches commonly use keyword matching and predefined skill dictionaries, whereas more advanced approaches use semantic representations and machine learning to understand contextual relationships between terms. The extracted requirements can then be converted into a structured representation that allows comparison with candidate profiles. In Career Compass AI, job-description analysis forms the second major input after resume processing, enabling the system to identify the requirements of the selected role before performing resume–job comparison.

Further, job-description analysis helps identify the relative importance of different requirements within a target role. Technical skills, frameworks, databases, cloud technologies and other qualifications can be separated into structured categories for easier comparison with candidate profiles. Semantic approaches are useful when equivalent skills are expressed using different terminology across job postings [5], [11]. Research on skill extraction from large job-posting datasets also shows that employer requirements can vary across roles and evolve over time [7]. It can improve the identification of both matching and missing competencies. In Career Compass AI, the extracted job requirements are used as the reference profile for skill-gap analysis. The identified missing skills can subsequently support project recommendations and personalized preparation activities. The job-description analysis therefore connects employer expectations with the student's existing technical profile. This enables the proposed system to move from simple job-description processing towards role-specific employability assessment.

2.4 RESUME–JOB MATCHING

Resume–job matching aims to determine how closely a candidate's profile corresponds to the requirements of a particular job. Existing research has investigated keyword-based matching, similarity measures, machine learning models, contextual embeddings and graph-based approaches for connecting candidate competencies with job requirements [1], [9], [14]. Simple keyword matching can provide useful results when the same terminology is used in both documents, but it may fail when equivalent skills are represented using different names. Semantic matching approaches attempt to address this limitation by considering the meaning and relationship between terms rather than relying only on exact string matches.

In such cases, normalization and semantic similarity can improve the identification of related skills even when exact terms are not identical. Research has also explored hybrid matching methods that combine rule-based features with semantic representations to improve candidate–job alignment [5], [11]. These methods can provide more meaningful matching results by considering both explicit skills and their contextual relationships. The matching process can therefore identify skills that are already available in a candidate profile as well as requirements that are not sufficiently represented. In Career Compass AI, the matching result is used to generate matching skills, missing skills and overall profile alignment for the selected role. The identified gaps are further used for project recommendations and role-specific assessment. Thus, resume–job matching acts as the central connection between candidate profiling and personalized career preparation.

2.5 SKILL GAP IDENTIFICATION AND RECOMMENDATION

Skill-gap identification involves determining the difference between the skills currently represented in a student's profile and the skills expected for a target job role. Research on skill recommendation systems has explored the use of resume information to identify missing competencies and recommend suitable skills or learning opportunities [5]. Other studies have examined job and course recommendation by considering resume semantics and individual career requirements [10], while research on labour-market skill extraction provides information about the competencies currently appearing in job postings [7]. These approaches help convert identified skill differences into actionable development priorities for students. The identified gaps can also be used to support personalized project recommendations and learning roadmaps. In Career Compass AI, skill-gap results are further connected with assessment and progress tracking to support continuous improvement.

2.6 AI-BASED ASSESSMENT AND LEARNING ANALYTICS

Identifying skills from a resume does not necessarily indicate the actual knowledge level of a student because a technology may be listed on a resume without sufficient practical understanding. AI-based assessment research has therefore explored the use of automated assessment, feedback and personalized learning approaches [19], [25]. Learning analytics provides another important component by using learner performance data to understand progress over time and support subsequent learning decisions [17]. Personalized learning-path research has similarly examined how learning resources can be arranged according to learner requirements and competency levels [21], [22].

These approaches can help determine whether the skills identified from a resume are supported by the student's demonstrated performance. Repeated assessments can provide additional evidence about learning progress and areas that require improvement [17]. Assessment results can also be used to prioritize specific skills in a personalized learning roadmap [21], [22]. In Career Compass AI, this information is connected with skill-gap analysis to provide more meaningful recommendations. Thus, the system considers both stated skills and demonstrated performance when supporting career-readiness evaluation.

2.7 LITERATURE SURVEY

S. No.	Research Area	Main Findings and Relevance to Proposed System
1	Resume Analysis using NLP	NLP techniques are used to extract skills, education, experience and other important information from resumes. This supports the resume analysis module of Career Compass AI.
2	Skill Extraction	Research focuses on identifying technical and implicit skills from candidate profiles. This supports skill identification and

		normalization in the proposed system.
3	Resume Recommendation	AI-based approaches provide resume analysis and . This supports resume quality evaluation and improvement guidance.
4	Skill Recommendation	Skill recommendation systems identify missing or relevant skills and suggest areas for improvement. This supports the skill-gap analysis module.
5	Job-Market Skill Analysis	Research analyses skills appearing in job postings to understand employer requirements.
6	Resume–Job Matching	Existing approaches compare candidate profiles with job requirements using keyword, similarity and semantic-based methods. This supports Career Compass AI.
7	Job and Course Recommendation	Resume semantics can be used to recommend suitable jobs and learning opportunities. This supports job-role and learning recommendations.
8	Skill Matching using NLP	NLP-based approaches identify relationships between resume skills and job requirements. This supports matching and missing-skill identification.
9	Learning Analytics	This supports skill progress tracking.

10	AI-Based Assessment	AI-based assessment approaches support automated evaluation of learner knowledge and performance. This provides the foundation for the mock assessment module.
11	Personalized Learning	Personalized learning systems generate learning paths according to individual learner requirements and competency levels. This supports personalized roadmap generation.
12	Automated Feedback	Automated feedback approaches provide learners with guidance based on assessment performance. This supports continuous improvement and roadmap updates.

Table 2.1 — Literature Survey

2.8 RESEARCH GAP

The literature indicates that considerable research has been conducted on resume analysis, skill extraction, resume–job matching, skill recommendation, learning analytics and personalized learning, but these capabilities are frequently treated as separate functions. Resume analysis can identify candidate information, matching systems can estimate alignment with a job, and recommendation systems can suggest additional skills or learning resources [1], [5], [10]. However, there is a need for a continuous workflow that connects the result of resume–job comparison with practical skill development, assessment and progress tracking.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 EXISTING SYSTEM

Existing systems for resume and career analysis mainly focus on resume parsing, keyword extraction, job-description analysis and candidate–job matching. A typical resume screening system accepts a resume as input, extracts relevant information such as education, experience and technical skills, and compares these details with the requirements of a job description[12]. Some systems use keyword-based techniques, while advanced approaches use machine learning, semantic similarity and contextual representations to improve matching performance. Skill recommendation systems extend this process by identifying additional skills that may be useful for a candidate, and some platforms provide job or course recommendations based on the candidate profile.

However, these functions are commonly provided as separate capabilities and may not form a continuous student-development workflow. In particular, a conventional resume analysis system generally provides a score or matching result without explaining how the student can improve the identified gaps. It may not connect missing skills with practical projects, validate the student's actual knowledge through role-specific assessments, track improvement across multiple attempts or generate an adaptive learning roadmap[15],[16]. Therefore, the existing approach is mainly focused on analysis and matching, whereas Career Compass AI extends the process towards analysis, improvement, assessment and continuous career preparation

S.No.	Existing System	Proposed Career Compass AI
1	Focuses mainly on resume analysis and job matching	Integrates resume analysis, job matching and career preparation
2	Uses resume and job-description information	Uses resume, job requirements and assessment performance
3	Provides matching	Identifies matching, missing and improvement skills

	or compatibility	
4	Recommendations may be generic	Recommends practical projects based on identified skill gaps
5	Limited validation of actual technical knowledge	Provides role-specific mock assessments
6	Usually provides a one-time analysis	Tracks performance across multiple assessment attempts
7	Limited learner progress tracking	Provides skill-wise progress analytics
8	Does not continuously update learning direction	Generates a personalized roadmap based on skill gaps and performance
9	Job recommendations may be separate from skill analysis	Connects suitable job roles with the student's skill profile
10	Mainly supports recruitment-oriented analysis	Focuses on student employability and continuous career preparation

Table3.1 — Existing and proposed system comparison

3.2 PROPOSED SYSTEM

Career Compass AI is proposed as an integrated AI and NLP-based employability assessment and career-readiness platform designed primarily for college students and fresh graduates. The system begins by accepting a student's resume and a target job role or job description as inputs[9]. The resume is processed to extract relevant information such as technical skills, education, projects, experience and other profile details, while the job description is analysed to identify required and preferred

skills, technologies and qualifications. These two structured profiles are then compared to determine matching skills, missing skills and areas that require improvement. Based on the identified gaps, the system recommends suitable practical projects that can help the student develop multiple required skills together[20].

The system also provides suitable job-role recommendations based on the student's profile and conducts a role-specific mock assessment to evaluate actual technical understanding. Assessment results are recorded at skill level and can be compared across repeated attempts to identify improvement or remaining weaknesses. The collected information is then used to generate a personalized roadmap containing prioritized learning areas and practical activities. Finally, the system presents a career-readiness indicator based on profile alignment, skill coverage and assessment performance. The proposed architecture therefore connects the different stages into a continuous process rather than treating resume analysis as the final output[25].

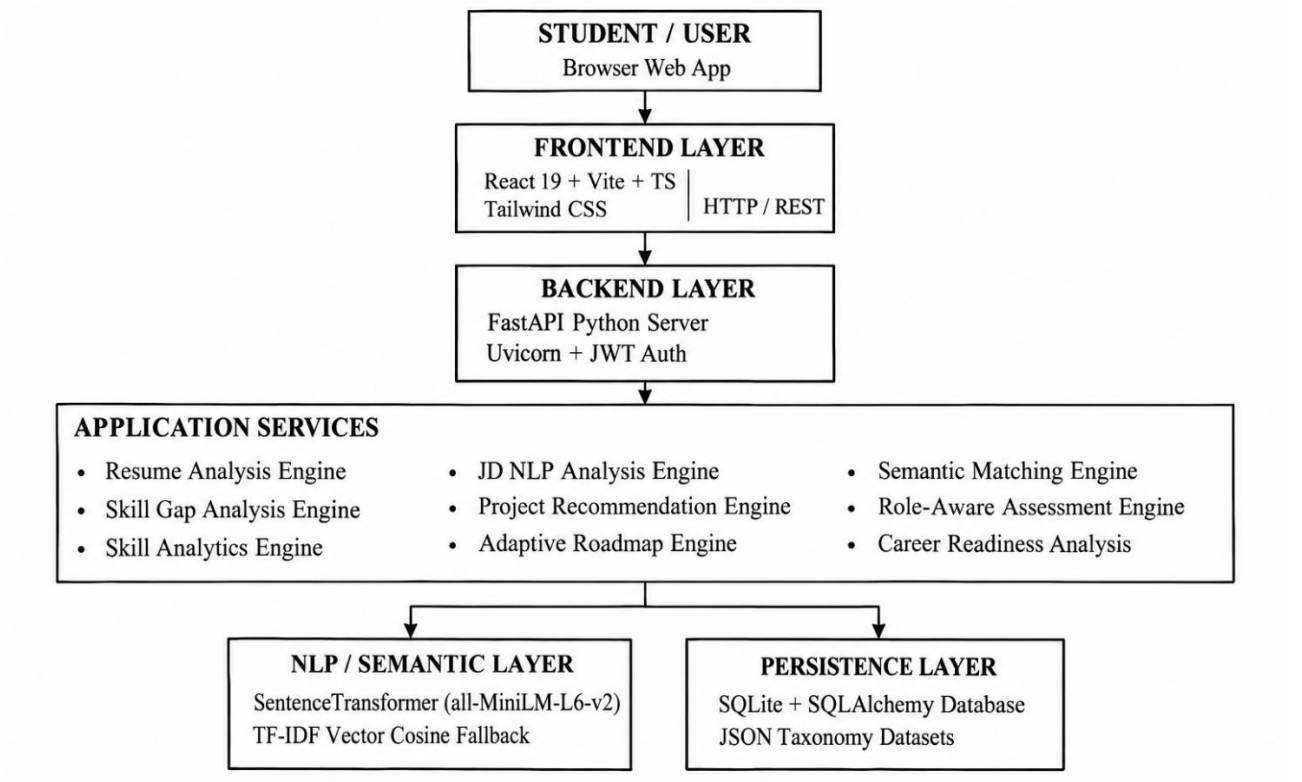


Figure 3.1. Overall Architecture of the Proposed System

3.3 SYSTEM WORKFLOW

The complete workflow of Career Compass AI follows a sequential but continuous process beginning with user authentication and ending with career-readiness analysis. After logging in, the student uploads a resume, which the system parses and analyses to identify key profile . The student then provides a target job description or selects a relevant job role[3]. The system analyses the job requirements and compares them with the skills extracted from the resume. The comparison produces matching skills, missing skills and improvement areas, with the identified gaps categorized according to their priority.

These results are passed to the recommendation module, which identifies practical projects capable of addressing the student's important missing skills and also determines suitable job roles based on the current profile. The student can then take a mock assessment containing questions related to the target role and identified skill areas[10]. The assessment engine evaluates the answers and generates skill-wise performance information the system compares the new performance with previous attempts and displays the progress through analytics. The updated skill information and assessment performance are then used to generate a personalized roadmap that prioritizes the remaining gaps. Finally, the system calculates a career-readiness indicator based on the defined profile, skill and assessment factors.

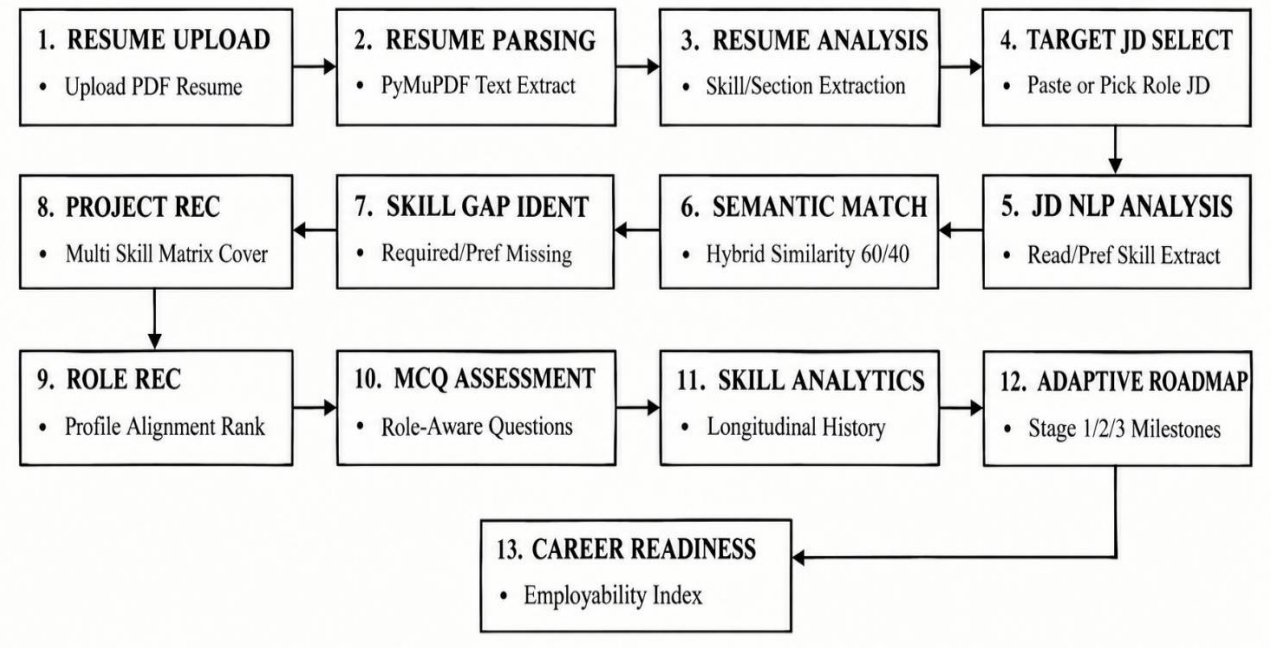


Figure 3.2— Complete Career Compass AI Workflow

CHAPTER 4

SYSTEM SPECIFICATION

4.1 HARDWARE REQUIREMENTS

Career Compass AI does not require high-end hardware for normal operation. A standard computer or laptop with a modern processor, minimum 8 GB RAM and sufficient storage is adequate for development and demonstration. A stable internet connection is required for accessing the web application and related services when necessary[5]. The system can be developed and tested using commonly available software and development tools without requiring specialized computing infrastructure. A modern web browser is sufficient for accessing the frontend interface and interacting with the platform. The backend services can run locally on a standard development environment with the required Python dependencies installed. Adequate storage is needed for maintaining uploaded resumes, application data and generated analysis results. The system should also have sufficient memory to support PDF processing and NLP-based analysis during execution. These requirements make the proposed platform suitable for student-level development, testing and demonstration environments.

Component	Minimum Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or equivalent
RAM	8 GB
Storage	256 GB SSD or higher
Display	1366 × 768 resolution or higher
Network	Stable Internet Connection
Input Devices	Keyboard and Mouse

Table 4.1 Hardware Requirements

4.2 SOFTWARE REQUIREMENTS

The system uses modern web development and AI/NLP technologies. React.js with Vite is used for the frontend, while Python-based technologies are suitable for backend and NLP processing. PostgreSQL can be used for persistent data storage, and supporting libraries are used for PDF processing, skill extraction, analysis and visualization[13].

Software / Technology	Purpose
Windows / Linux	Development and execution environment
React.js	Frontend development
Vite	Frontend build and development tool
TypeScript	Frontend programming
Tailwind CSS	User interface styling
Python	Backend and AI/NLP processing
FastAPI	Backend API development
PostgreSQL	Database management
PDF.js / PyMuPDF	PDF text extraction
spaCy	NLP and skill processing
scikit-learn	Machine learning and similarity processing
Recharts	Data visualization
Git / GitHub	Version control and project management

Table 4.2 Software Requirements

4.3 DATASET SPECIFICATION

The system processes several types of data during its operation. The main inputs are student resume documents and job descriptions. Resume data includes personal profile information, education, technical skills, projects, certifications and experience, while job-description data contains role requirements, skills, technologies and qualifications. Assessment data includes questions, answers, scores and skill-wise performance[5][14]. The system also maintains recommendation and roadmap information to support personalized career preparation. The collected data is processed through different stages depending on the requirements of each module. Resume information is first extracted and organized into structured fields for further analysis. Job-description information is similarly processed to identify the skills and qualifications associated with the selected role.

The structured data is then used for resume–job comparison and skill-gap identification. Matching results contain the skills already represented in the candidate profile as well as the skills that are not sufficiently covered. Project and job-role recommendation data is generated based on the identified requirements and skill gaps. Assessment records are maintained to track individual attempts and performance across different skill areas. Roadmap information is linked with the identified improvement areas to provide a structured preparation sequence. The system can therefore connect data from resume analysis, job matching, assessment and learning guidance within a common workflow. This integrated data handling supports continuous career preparation rather than a single resume-based evaluation.

4.4 FUNCTIONAL REQUIREMENTS

The system shall allow users to register and log in securely, upload and analyse resumes, provide job descriptions, identify matching and missing skills, recommend suitable projects and job roles, conduct mock assessments, display skill-wise performance, track improvement across attempts and generate a personalized learning roadmap. The system shall also provide a career-readiness indicator based on the defined scoring components[8],[18]. The system shall maintain assessment history and user-specific progress information for future review. It shall also use the identified skill gaps and assessment performance to support personalized recommendations and roadmap generation. These outputs shall help users continuously monitor and improve their career preparation for the selected role.

Requirement	Description
User Authentication	Allows secure registration and login
Resume Analysis	Extracts and analyses information from resumes
JD Analysis	Identifies job requirements and relevant skills
Skill Matching	Compares resume skills with job requirements
Skill Gap Analysis	Identifies missing and improvement skills
Project Recommendation	Recommends projects based on skill gaps
Job Role Recommendation	Suggests suitable roles based on the profile
Mock Assessment	Evaluates technical knowledge through questions
Skill Analytics	Tracks skill performance and improvement
Personalized Roadmap	Provides prioritized learning activities
Readiness Indicator	Displays overall career-preparation status

Table 4.3 Functional Requirements

4.5 NON-FUNCTIONAL REQUIREMENTS

The system should provide a simple and understandable user interface so that students can easily navigate between analysis, recommendations, assessment and roadmap modules. It should provide reasonable response time for normal operations, maintain user-data security, validate uploaded files and inputs, and ensure that one user's information is not accessible to another user[14]. The system should also be

maintainable and scalable so that additional skills, job roles, assessment questions and recommendation rules can be added without redesigning the complete application[2].

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4.6 DEVELOPMENT ENVIRONMENT

The system is developed using a web-based architecture with React.js and Vite for the frontend and Python-based technologies for backend processing and AI/NLP functionality. The application uses structured data processing for resume and job-description analysis, while PostgreSQL is considered for persistent storage of user profiles, resumes, job descriptions, assessment attempts, skill progress and roadmap information. Git and GitHub are used for version control, and modern browsers such as Google Chrome, Microsoft Edge or Mozilla Firefox can be used to access the application[14-21].

The architecture also supports modular integration between the frontend, backend services and data-processing components. The backend manages application requests and coordinates resume processing, job-description analysis, skill matching, assessment and roadmap generation. NLP and supporting libraries are used to extract and process relevant information from uploaded documents. The system maintains structured records so that analysis results and user progress can be retrieved when required. This architecture allows individual modules to be updated or extended without significantly affecting the overall application. It also provides a suitable foundation for integrating additional AI-based services and career-preparation features in future versions.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 SYSTEM OVERVIEW

Career Compass AI is implemented as a modular web-based application that connects resume processing, job-description analysis, skill matching, recommendation, assessment and career planning. The system follows a phased implementation approach so that the output of one module can be used by the next module. The frontend provides the user interface, while analysis and recommendation modules process the collected information and generate meaningful results. The major implementation phases are authentication, resume analysis, job-description analysis, resume–job matching, skill-gap identification, project and job-role recommendation, mock assessment, skill analytics, personalized roadmap and readiness analysis.

The implementation is organized into independent modules to simplify development, testing and maintenance. The authentication module manages user registration, login and protected access to user-specific information. The resume analysis module extracts relevant content from uploaded documents and identifies important candidate skills and profile details. The job-description module processes role requirements and prepares the information required for comparison. The matching module compares resume skills with the selected job requirements and identifies matching and missing competencies. These results are then passed to the recommendation module for selecting relevant projects and job roles. The assessment module uses the selected role and relevant skills to evaluate the student's technical understanding. Assessment results are stored and used by the analytics and roadmap modules to identify improvement areas. Finally, the readiness analysis combines the defined outputs to provide an overall indication of the student's current preparation status.

5.2 SYSTEM DESIGN

The system is designed using separate functional modules to simplify development and maintenance. User information and uploaded documents are processed first, followed by skill extraction and comparison with job requirements. The identified skill gaps are passed to recommendation and assessment modules.

Assessment performance is then used by the analytics and roadmap modules. This modular design allows individual components to be improved or extended without changing the complete application.

The modular approach also improves the reliability of data flow between different stages of the application. Each module performs a specific function and passes its processed output to the next relevant stage. This reduces dependency between individual components and makes debugging easier during development. It also allows new job roles, skills, assessment questions and recommendation rules to be added with minimal changes to the existing modules. The overall design therefore supports structured implementation while maintaining flexibility for future enhancements.

5.3 MODULE-1 USER AUTHENTICATION

PhaseThe authentication module provides controlled access to the Career Compass AI platform. A user can register using basic details and log in using valid credentials. Authentication ensures that user-specific resume, job-description and assessment information can be associated with the correct account. The interface also provides profile information and logout functionality.

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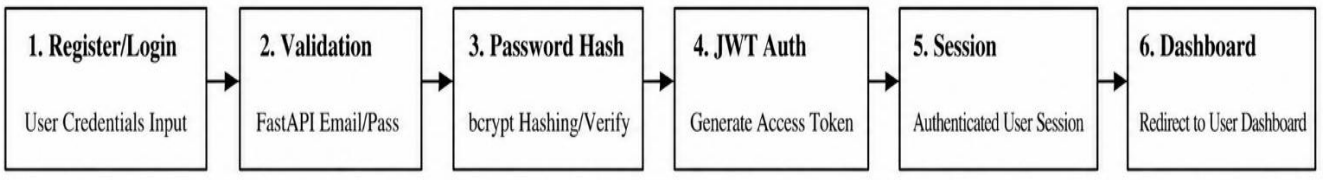


Figure 5.1. User Authentication Workflow

5.4 MODULE-2 RESUME UPLOAD AND ANALYSIS

The resume module allows the student to upload a resume document, particularly in PDF format. The uploaded document is processed to extract its text, after which relevant information such as technical skills, education, experience, projects and other profile details is identified. The extracted information is analysed using predefined skill data and resume-analysis rules. The module produces indicators such as resume quality, ATS-oriented score, skill coverage and domain fit.

The extracted resume data is converted into a structured form so that it can be reused by the subsequent modules. The identified skills are normalized to improve consistency when different names or abbreviations are used for the same technology. The analysed resume information is then made available for comparison with the requirements of a selected job role. The module also highlights strengths and areas that may require improvement in the current resume profile. These outputs form the initial input for resume–job matching, skill-gap identification and personalized career recommendations.

Component	Purpose
PDF Processing	Extracts text from the uploaded resume
Text Processing	Cleans and prepares extracted content
Skill Extraction	Identifies technical and relevant skills
Section Analysis	Checks important resume sections
Keyword Analysis	Measures relevant keyword coverage
Resume Scoring	Generates resume-quality indicators
Domain Analysis	Identifies possible technical domain alignment

Table 5.1 – Resume analysis components

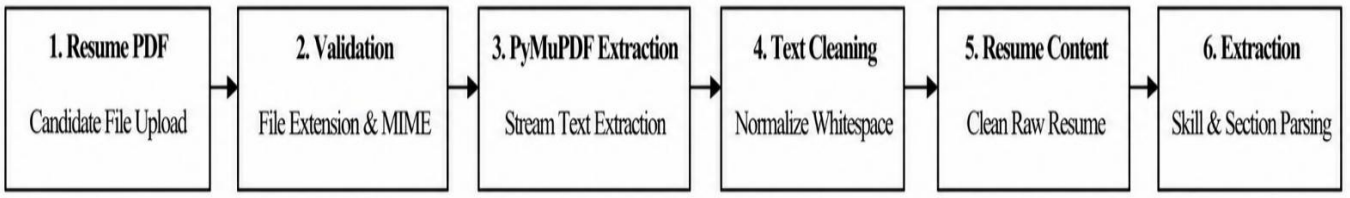


Figure 5.2. Resume upload and parsing workflow

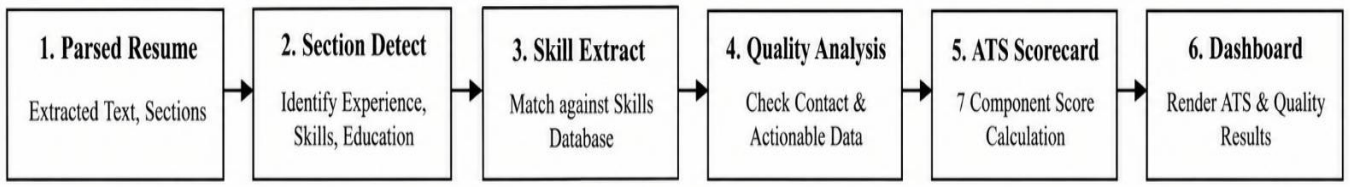


Figure 5.3. Resume analysis workflow

5.5 MODULE-3 JOB DESCRIPTION ANALYSIS

The job-description module accepts a target job description or role information from the student. The system analyses the provided content and identifies relevant skills, technologies and requirements. Required and preferred skills are separated where possible, and the extracted information is structured for comparison with the student's resume profile. This module provides the input required for the subsequent matching and skill-gap stages.

The extracted requirements are organized into a structured format to support consistent comparison with the resume data. Skill names are normalized so that equivalent technologies or common variations can be identified during matching. The module also provides the selected role requirements as the reference for determining the student's matching and missing skills. These results are passed to the skill-gap module for further analysis. The structured job information also supports the generation of role-specific projects, assessments and career-preparation recommendations.

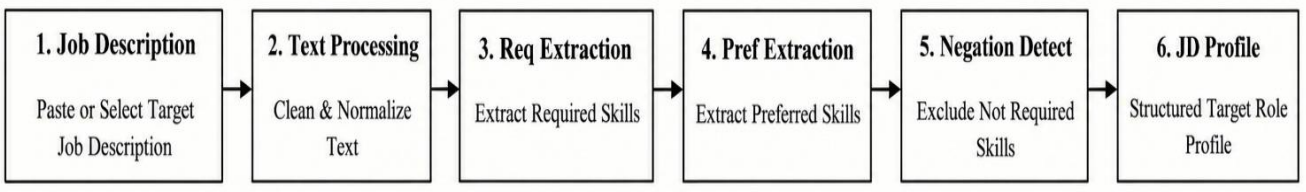


Figure 5.4. Job description analysis workflow

5.6 MODULE 4 – RESUME–JOB MATCHING AND SKILL GAP

The resume–job matching module compares the skills extracted from the student's resume with the skills identified from the selected job description. Skills present in both profiles are treated as matching skills, while requirements not represented in the resume are identified as missing skills. The system then assigns priorities to important gaps so that students can focus on areas requiring greater improvement. The matching process considers the selected job role as the reference profile for evaluating candidate alignment. Skill names are normalized before comparison to handle common variations, abbreviations and equivalent technology names. The system can therefore identify relevant matches even when the same skill is represented using slightly different terminology.

The comparison result provides a clear view of the competencies already covered by the student. It also highlights the specific requirements that are not sufficiently represented in the resume. These missing skills are used as the basis for generating relevant project recommendations. The matching results can also support the selection of role-specific assessment questions. The identified gaps are further passed to the roadmap module for focused skill development. This creates a direct connection between job requirements, skill-gap identification and subsequent career-preparation activities. Thus, the module converts the resume and job description into an actionable skill-level comparison for the student.

Priority	General Meaning
High	Important skill required for the target role and currently missing or weak
Medium	Relevant skill that can improve role alignment
Low	Additional skill useful for profile improvement

Table 5.2 Skill gap identification workflow

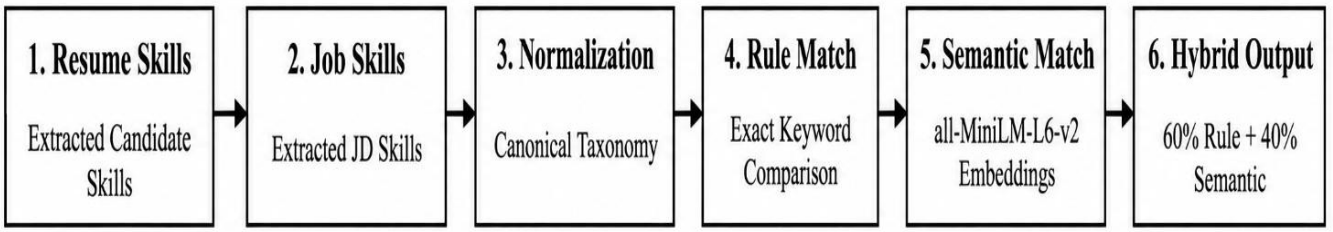


Figure 5.5. Resume – Job Skill Matching Workflow

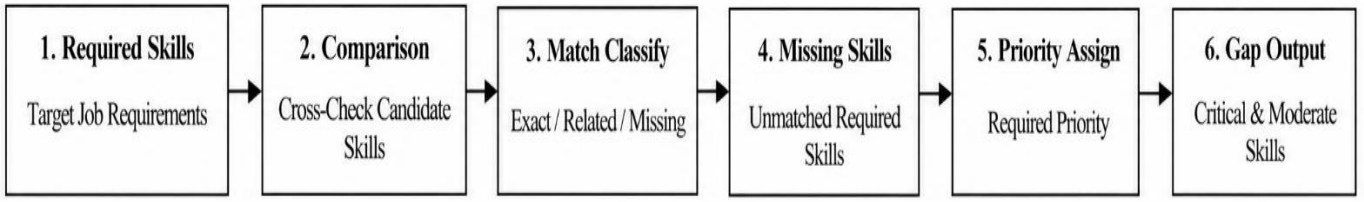


Figure 5.6. Skill Gap Identification Workflow

5.7 MODULE 5 – PROJECT AND JOB ROLE RECOMMENDATION

The recommendation module uses the identified skill gaps to suggest practical projects and suitable job roles. Projects are mapped to the skills they help develop, allowing one project to address multiple missing skills. For example, a project involving Spring Boot, MySQL, REST APIs and Docker can help a student improve several backend-related requirements together. Job-role recommendations are generated by comparing the student's current skill profile with predefined role requirements[15]. The recommended projects are selected based on the specific skills identified as missing from the student's profile.

This makes the recommendations more relevant than providing a general list of projects. Each project can be associated with multiple technologies so that students can address several skill gaps through practical implementation. The job-role recommendation component similarly evaluates the student's current competencies against the requirements of available roles. The system presents roles that have stronger alignment with the student's existing skills. It also helps students understand which additional skills may be required to become better prepared for a selected role. The recommendations can therefore support both immediate project selection and longer-term career planning.

These outputs are connected with the assessment module so that students can further evaluate their understanding of the recommended skill areas. The roadmap module can also use the identified gaps to organize suitable learning priorities. Overall, the recommendation module converts the results of skill-gap analysis into practical and role-oriented career preparation activities.

Missing Skills	Example Recommended Project	Skills Covered
Spring Boot, REST API, MySQL, Docker	Employee Management System	Spring Boot, REST API, MySQL, Docker
Python, SQL, Data Analysis	Sales Analytics Dashboard	Python, SQL, Data Analysis
React, JavaScript, API Integration	Job Portal Interface	React, JavaScript, APIs
Machine Learning, Python, Statistics	Student Performance Prediction	Python, ML, Statistics

Table 5.3- Project Recommendation Mapping

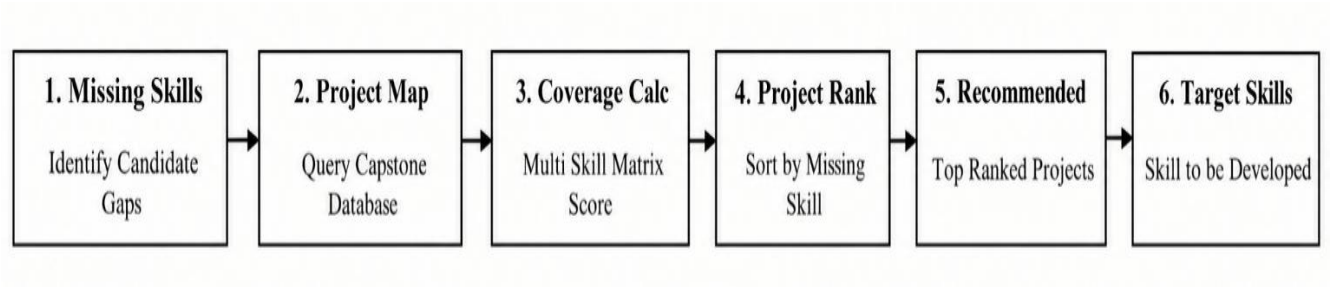


Figure 5.7 - Project Recommendation Workflow

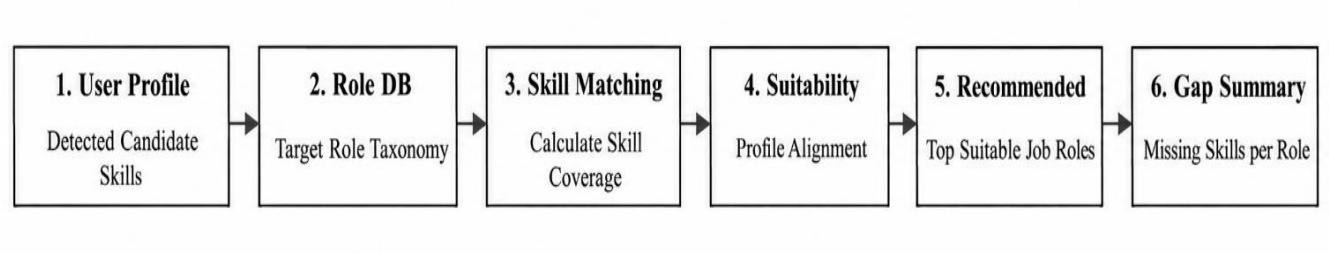


Figure 5.8. Job Role Recommendation Workflow

5.8 MODULE 6 – AI MOCK ASSESSMENT

The mock assessment module evaluates the student's understanding of skills relevant to the selected job role. Questions are associated with technical skills and difficulty levels, and the student's answers are evaluated to calculate an overall assessment score and skill-wise performance. The assessment provides an additional validation of technical knowledge because the presence of a skill in a resume does not necessarily indicate the student's actual proficiency. The assessment questions are selected according to the skills associated with the chosen role and the student's identified profile. The module evaluates different technical areas so that performance can be examined at a skill-wise level.

After submission, the system records the student's answers and calculates the corresponding score. The results help identify skills in which the student demonstrates stronger understanding and areas where further preparation is required. Assessment performance is stored so that previous attempts can be reviewed and compared. Repeated assessments allow students to observe whether their performance improves after completing recommended learning activities. The identified weak areas can also be passed to the analytics and roadmap modules for further skill development. This creates a connection between skill-gap identification, knowledge validation and personalized preparation. Thus, the mock assessment provides an additional evidence-based component for evaluating the student's career readiness.

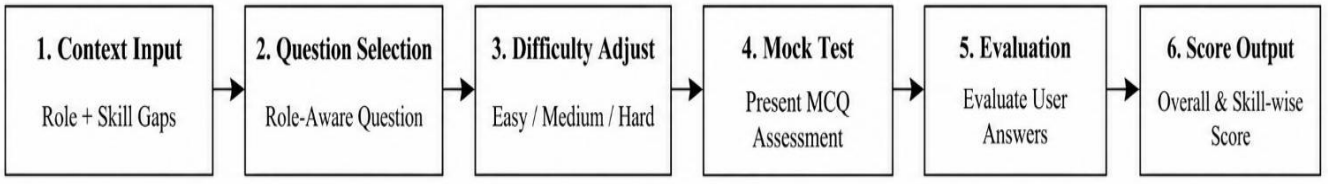


Figure 5.9 – AI Mock Assessment Workflow

5.9 MODULE 7 – SKILL ANALYTICS AND PROGRESS TRACKING

The skill analytics module stores assessment performance and compares results across multiple attempts. Instead of replacing previous results, each assessment attempt can be considered separately so that changes in performance can be observed. Skill-wise scores and improvement values can be represented using charts, allowing students to identify stronger areas and skills that still require attention. The analytics module provides a visual representation of assessment progress so that students can understand their development more easily.

Performance trends can be reviewed for individual skills as well as overall assessment results. The module helps identify consistently weak areas that may require additional preparation. It also allows students to compare recent performance with earlier attempts and observe improvement over time. These insights are used as supporting information for generating and updating the personalized learning roadmap. Thus, skill analytics connects assessment history with continuous skill development and career preparation.

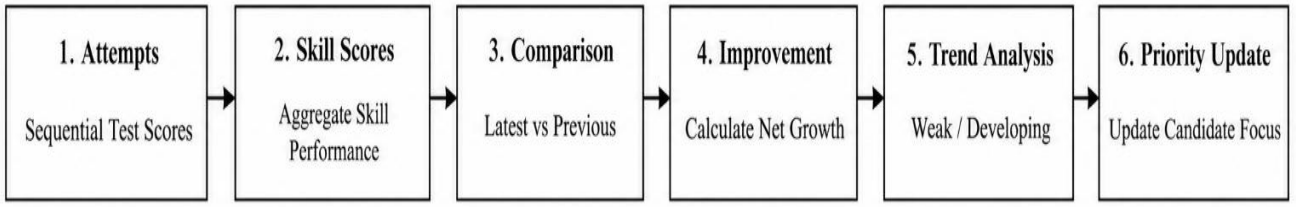


Figure 5.10 – Skill Analytics And Progress Tracking

5.10 MODULE 8 – PERSONALIZED ROADMAP AND READINESS

The roadmap module converts the identified skill gaps and assessment performance into prioritized learning activities. Skills with greater importance and weaker assessment performance can receive higher priority. The final readiness indicator combines profile alignment, skill coverage and assessment performance according to the defined scoring framework. It is intended as a preparation indicator rather than a guarantee of job selection. The roadmap module converts the identified skill gaps and assessment performance into prioritized learning activities. The final readiness indicator combines profile alignment, skill coverage and assessment performance according to the defined scoring framework. It is intended as a preparation indicator rather than a guarantee of job selection.

The roadmap provides a structured sequence of skills that the student can focus on based on the current analysis results. It helps students understand which areas should be improved first and which skills can be addressed at later stages. The roadmap can consider both missing skills from the selected job role and weak areas identified through the mock assessment. This allows the learning priorities to be connected directly with the student's current performance. The recommended activities can therefore provide a clear direction for further preparation. As the student completes learning activities and additional assessments, the roadmap can be reviewed to identify remaining areas of improvement. Assessment history also provides supporting information for understanding changes in skill performance over time. The readiness indicator presents the combined outcome of the major analysis stages in a simple form. It helps students understand their current level of preparation for the selected role. The indicator can be used as a reference point before and after improving the identified skill gaps. Overall, the roadmap and readiness modules provide the final stage of the Career Compass AI workflow by connecting analysis results with continuous career preparation.

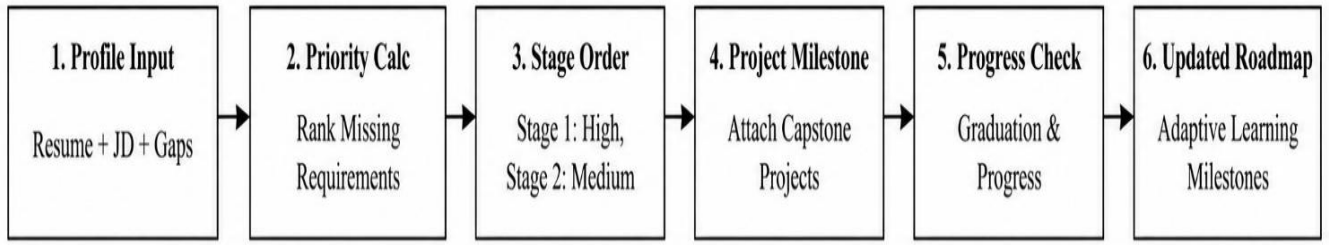


Figure 5.11 – Personalized Roadmap Workflow

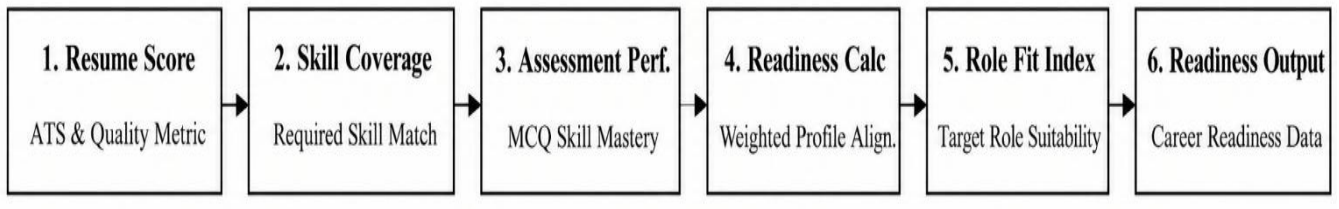


Figure 5.12 – Career Readiness Indicator

5.11 TECHNOLOGY STACK

The implementation uses React.js, Vite, TypeScript and Tailwind CSS for the frontend interface, with supporting libraries such as Zustand and Recharts for state management and data visualization. Python and FastAPI are used for backend-oriented processing, while PostgreSQL is used as the planned persistent database layer. PDF processing, NLP libraries and structured skill datasets support resume and job-description analysis. Git and GitHub are used for version control and project management. The technology stack is selected to support modular development, responsive visualization and future integration of advanced NLP and AI capabilities. The frontend components are organized into separate modules so that each major function can be developed and maintained independently.

React.js is used to create reusable interface components for resume upload, job analysis, skill-gap analysis, assessment and roadmap visualization. Vite provides a lightweight development environment and supports efficient frontend builds. TypeScript improves code reliability by providing structured type definitions for application data and component properties. Tailwind CSS is used to maintain a consistent and responsive interface across different screen sizes. Zustand manages shared application state required for user information, resume analysis and matching results. Recharts is used to present assessment performance and skill-wise progress through graphical visualizations.

On the backend, FastAPI provides API endpoints for communication between the frontend and processing services. Python-based modules handle document processing, skill extraction, matching and recommendation logic. PDF processing libraries are used to extract text from uploaded resume documents before analysis. NLP components support skill identification and comparison between resume and job-description information. Structured skill data provides a consistent reference for matching and recommendation operations. Git and GitHub support version tracking, code management and collaborative development throughout the project. The modular technology stack also allows additional AI, NLP and cloud-based services to be integrated in future versions.

CHAPTER - 6

RESULT AND DISCUSSION

6.1 USER AUTHENTICATION RESULTS

The Career Compass AI system successfully implemented user authentication as an essential component for controlling access to personalized career assessment features. The authentication workflow provides a structured mechanism through which users can create an account, log into the application, maintain an authenticated session, access protected modules, and securely log out from the system. Authentication is integrated with the frontend and backend so that user-related operations are performed only after successful verification.

Once a user is authenticated, the system maintains the corresponding authentication state and allows access to the modules required for resume analysis and career readiness assessment. This authentication mechanism also supports user-specific data handling, ensuring that resume information, job descriptions, assessment attempts, analytics, roadmap information, and readiness results are associated with the appropriate user account. The authentication workflow was validated as part of the complete system integration process, and the results confirmed that registration, login, session handling, protected access, and logout operate as connected components of the application.

The registration process allows new users to create their individual accounts by providing the required information. During login, the entered credentials are verified before access to the application is granted. The authenticated session is then used to maintain the user's access state while navigating between protected modules. This ensures that users can continue their analysis workflow without repeatedly providing authentication details. Protected routes prevent unauthorized users from directly accessing user-specific application pages.

The system also handles invalid login credentials and prevents access when authentication requirements are not satisfied. Logout functionality terminates the active session and redirects the user away from protected application areas. The authentication mechanism therefore provides a basic security layer for the overall career assessment workflow. User-specific records can be retrieved and processed according to the authenticated account. This helps maintain separation between the data of different users during normal application operation.

The integration of authentication with the other modules also provides a consistent user experience throughout the platform. Overall, the implemented authentication workflow establishes the required access-control foundation for Career Compass AI and supports secure interaction with its personalized career-preparation features.

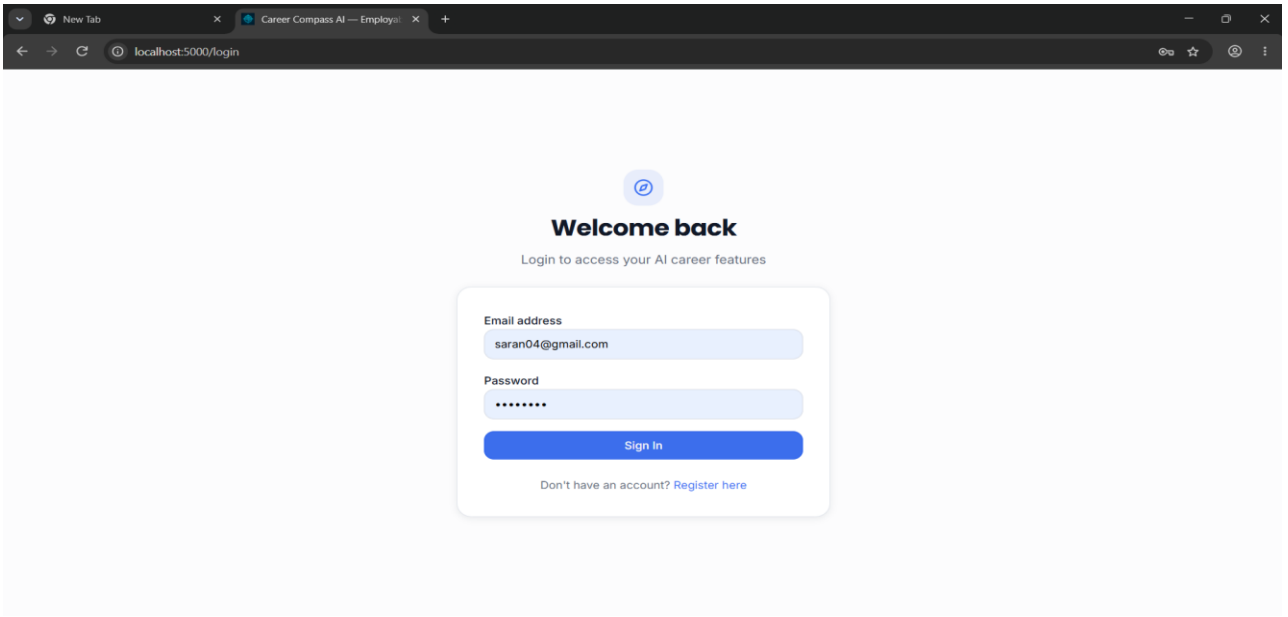


Fig. 6.1 – Successful Authentication Result

6.2 USER REGISTRATION RESULTS

The user registration module was developed to allow new users to create an account before accessing the personalized features of Career Compass AI. During registration, the user provides the required account information through the registration interface, and the submitted information is transferred from the React frontend to the FastAPI backend through the corresponding authentication API. The backend validates the registration request and creates the user record in the SQLite database through the SQLAlchemy database layer. Password information is handled through the implemented password hashing mechanism rather than being stored as plain text. The registration process was tested using valid user information to verify successful account creation and was also checked with duplicate registration attempts to ensure that an existing account could not be incorrectly created again.

After successful registration, the user can proceed to the login process using the registered credentials. The registration results demonstrated that the frontend form, authentication API, backend validation, password handling, and database persistence

are correctly connected, providing a functional account creation workflow for the Career Compass AI platform. The registration workflow also provides the initial association between the user account and the personalized data generated by the application. Once the account is created, the registered credentials can be used to authenticate the user during subsequent access.

The database layer maintains the required user information so that the account can be identified during later authentication requests. Validation at the backend level helps prevent incomplete or invalid registration requests from being processed. The duplicate-account check also helps maintain uniqueness of user records within the system. The password hashing mechanism provides an additional security measure by avoiding direct storage of the original password.

The registration API was verified for successful communication between the frontend and backend components. The database persistence was also checked to confirm that valid registration details were stored correctly. The workflow therefore establishes a reliable connection between the registration interface, authentication service and database layer. This account information is subsequently used by the login and session-management components. As a result, the registration module forms the first stage of the authenticated user workflow in Career Compass AI.

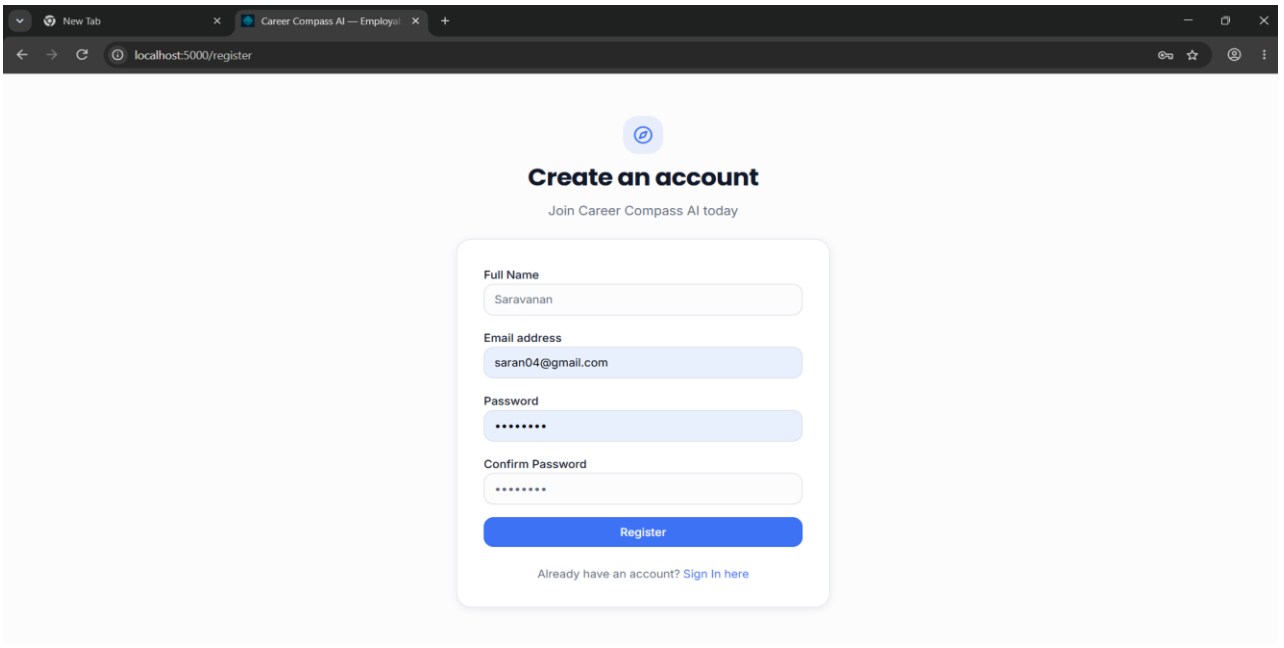


Figure 6.2 – Career Readiness Dashboard

6.3 USER LOGIN RESULTS

The user login module enables registered users to securely enter the Career Compass AI application and access their personalized career analysis workflow. During login, the user enters the credentials created during registration, and the frontend sends the authentication request to the FastAPI backend. The backend verifies the supplied credentials against the stored user information and, when the credentials are valid, generates an authentication token for the user session. The frontend stores the authenticated state and uses the authentication information when communicating with protected backend APIs.

Successful login allows the user to continue to the main application modules, including resume processing, resume analysis, job description analysis, skill-gap analysis, project recommendations, mock assessment, skill analytics, learning roadmap, and readiness analysis. Invalid login scenarios were also considered to ensure that incorrect credentials do not result in unauthorized access. The login results confirmed that the authentication API, credential verification, token generation, frontend authentication state, and protected application workflow are properly integrated and operate together as expected.

The login workflow also ensures that only registered users with valid credentials can enter the personalized application environment. After successful authentication, the generated token is used to maintain the user's authenticated session during subsequent requests. Protected API endpoints verify the authentication information before processing user-specific operations. This prevents unauthenticated requests from directly accessing protected career assessment features.

The frontend authentication state is updated after successful login so that the appropriate application modules become available to the user. The system also provides suitable handling for failed authentication attempts without creating an authenticated session. The login process was verified through valid and invalid credential scenarios to confirm the expected behavior. Successful authentication allows the user to move through the complete career-readiness workflow without requiring repeated login operations. The authenticated user context also supports the retrieval and storage of data associated with the corresponding account.

This provides continuity between resume analysis, job matching, assessment and roadmap activities. Therefore, the login module acts as an important connection between account authentication and the personalized features of Career Compass AI. Overall, the implemented workflow provides controlled access while maintaining the required integration between the frontend, backend authentication service and protected application modules.

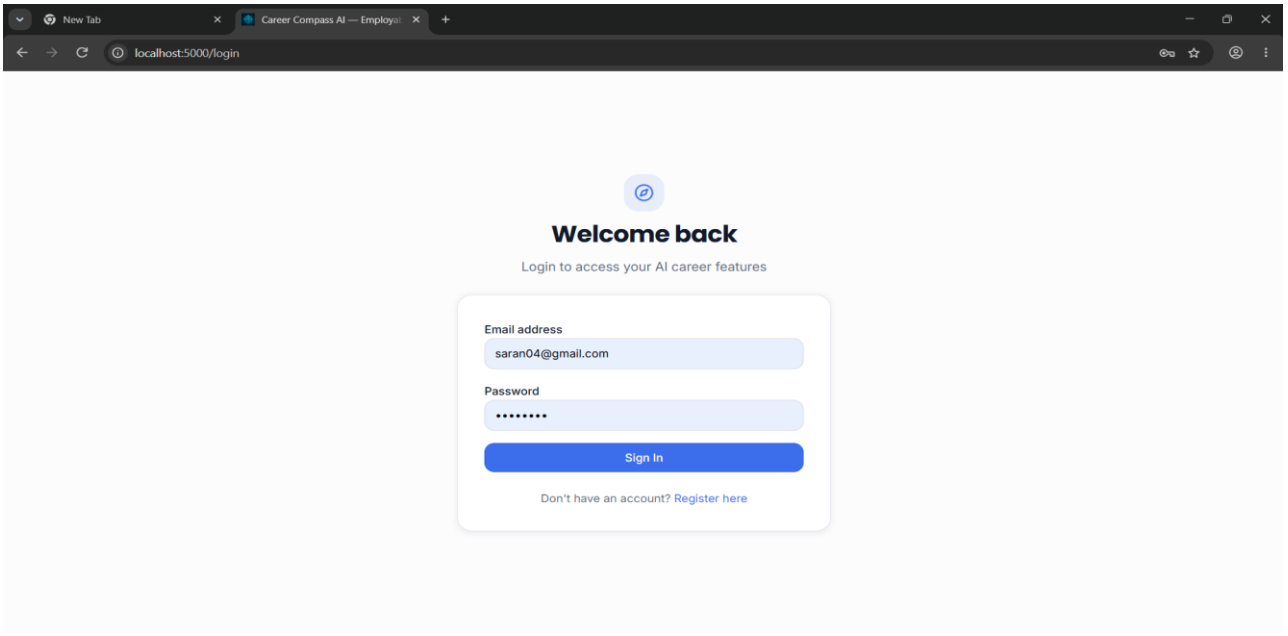


Fig 6.3 - User Login Results

6.4 AUTHENTICATION AND SESSION MANAGEMENT RESULTS

Authentication and session management provide continuity between the user's login process and subsequent interactions with the Career Compass AI platform. After successful login, the authenticated session is maintained so that the user can navigate between the different application modules without repeatedly entering credentials. The frontend authentication state is connected with the backend authorization mechanism, allowing protected API requests to be associated with the currently authenticated user.

This session state is particularly important because the application contains personalized information such as uploaded resumes, analyzed job descriptions, assessment attempts, skill analytics, roadmap progress, and readiness results. The system was validated by navigating between protected pages after login and verifying that the authenticated state remained available during normal application usage. The session behavior was also checked during logout and subsequent access attempts to

ensure that the authenticated state was correctly cleared. These results demonstrate that authentication and session management provide the required continuity for the complete Career Compass AI workflow while maintaining the connection between the authenticated user and their application data.

The session mechanism also helps maintain a consistent user context across the different stages of the application. Each protected request is processed based on the authenticated user information available to the backend. This allows the system to retrieve and manage user-specific records without mixing them with information belonging to other accounts. The authentication state remains active while the user moves between the analysis, assessment, analytics and roadmap modules. When the user performs a logout operation, the active authentication state is removed and access to protected resources is restricted.

Subsequent attempts to access protected modules without valid authentication are therefore handled as unauthorized requests. The session-management workflow was tested to confirm that authentication status was correctly maintained during normal navigation and cleared after logout. This validation also confirmed the connection between frontend session handling and backend authorization checks. Overall, the implemented mechanism provides a consistent authenticated experience and supports secure access to the personalized features of Career Compass AI.

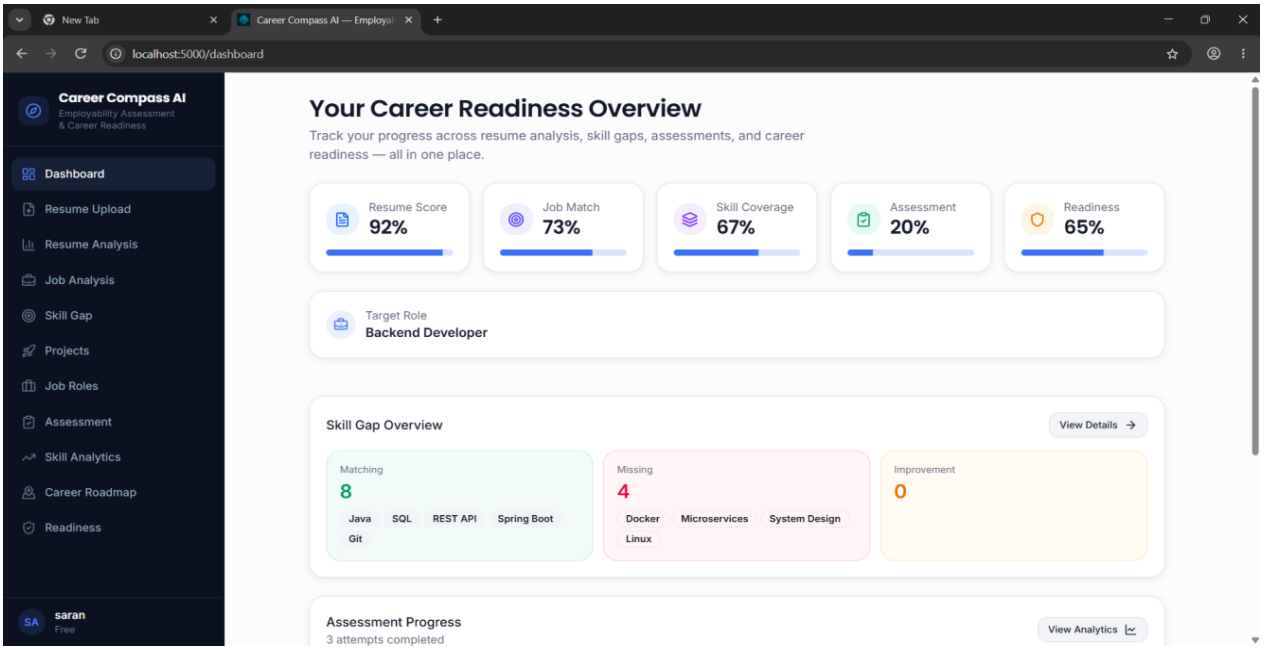


Fig. 6.4 – Authentication and Session Management Result

6.5 PROTECTED ACCESS AND LOGOUT RESULTS

Protected access and logout functionality were implemented to ensure that application features requiring authentication cannot be accessed directly by unauthenticated users. When an unauthenticated user attempts to access protected modules, the application checks the authentication state and prevents direct access until the user completes the login process. After successful authentication, the user can access the protected career-development workflow and use the available personalized modules. Logout functionality provides the reverse operation by terminating the active authenticated session and clearing the associated authentication state from the application. After logout, attempts to reopen protected pages or perform authenticated operations require the user to authenticate again.

This behavior was validated through protected-route navigation, authenticated API requests, logout, and subsequent access attempts. The results confirmed that the authentication lifecycle is completed correctly from registration and login through session management, protected resource access, and logout, thereby providing controlled access to the personalized features of the Career Compass AI system. The protected-access mechanism also helps maintain separation between authenticated and unauthenticated application states.

Before accessing protected modules, the system verifies whether a valid authentication state is available. This prevents users from bypassing the normal login workflow through direct page navigation. The same authentication requirement is applied when accessing backend operations that handle personalized user information. After logout, the previously active session information is cleared so that the application no longer treats the user as authenticated. The user must therefore complete the login process again before accessing protected features.

The workflow was tested using both authenticated and unauthenticated navigation scenarios. These tests confirmed that valid users could access the required modules while unauthorized access was appropriately restricted. Logout testing also confirmed that protected resources could not be accessed using the previous session state. This provides an additional access-control layer for the resume, assessment, analytics and roadmap information maintained by the system. Overall, protected access and logout complete the authentication lifecycle and support controlled interaction with the personalized Career Compass AI platform.

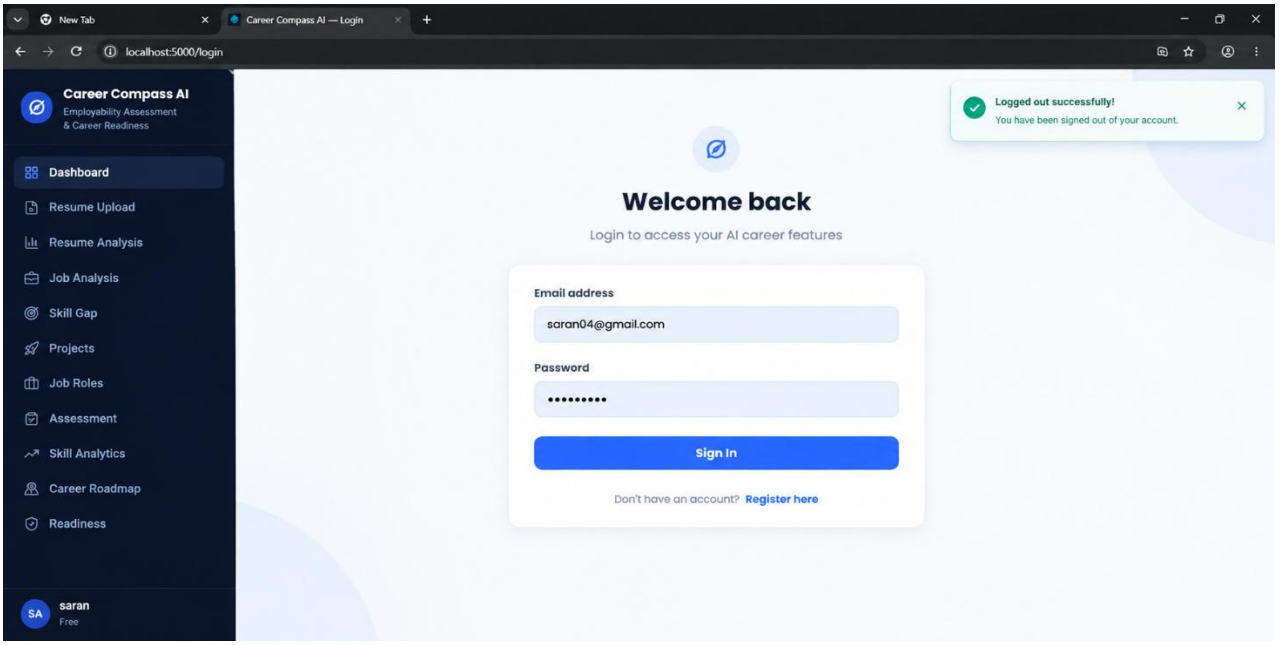


Fig. 6.5 – Protected Access and Logout Result

6.6 RESUME PROCESSING RESULTS

The resume processing module successfully provides the initial stage of the Career Compass AI workflow by accepting a resume document and extracting its textual content for further analysis. The extracted information is used to identify relevant resume sections, technical skills and other profile details. The processing stage converts the uploaded document into structured information that can be used by the subsequent analysis and matching modules. The uploaded resume is first validated to ensure that the provided document contains sufficient information for further processing. The text extraction stage handles the resume content and prepares it for subsequent skill identification and profile analysis.

Relevant information such as technical skills, education, experience and project details can then be organized into a structured representation. Skill names are normalized during processing to improve consistency when different naming formats or common abbreviations are used. The processed information is passed to the resume analysis module for calculating the required profile indicators. It also provides the resume data required for comparison with a selected job description. The extracted skills therefore act as an important input for identifying matching and missing competencies. The processing stage was tested using valid resume documents to verify that the extracted content could be used by the following modules.

Invalid or insufficient documents are handled through the implemented validation mechanism before analysis is performed. This ensures that subsequent modules receive meaningful resume information for processing. Overall, the resume processing module establishes the connection between the uploaded document and the remaining Career Compass AI analysis workflow.

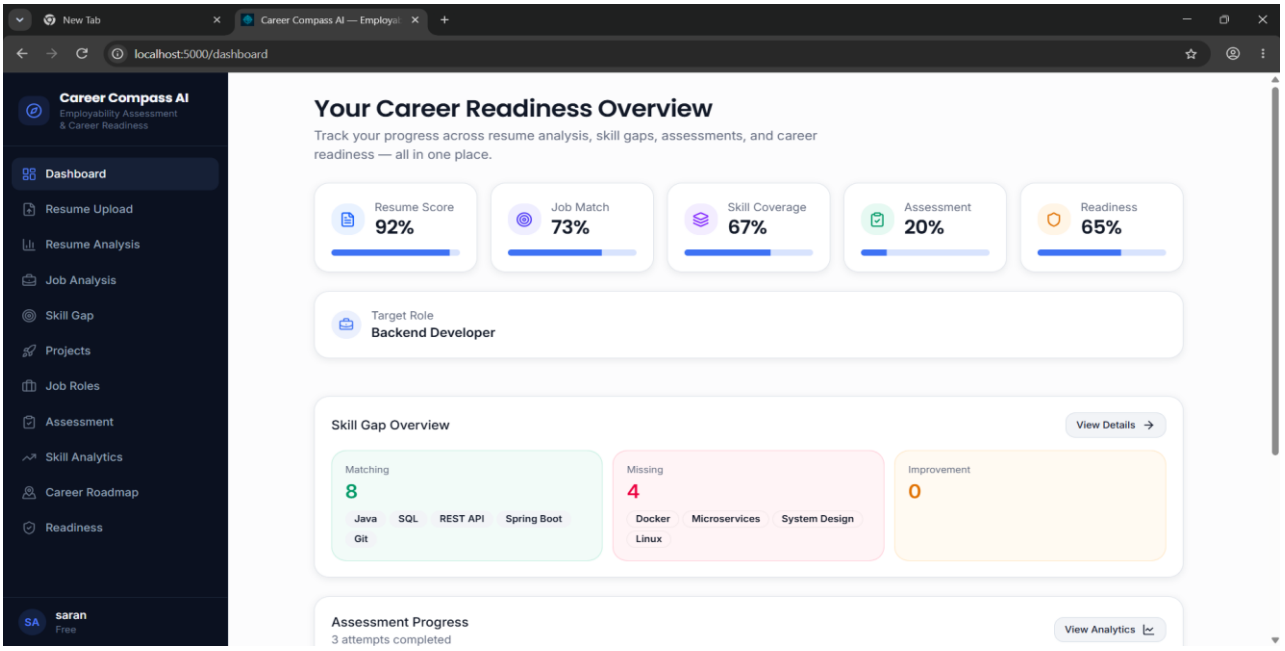


Figure 6.6 – Resume Upload And Processing Output

6.7 TEMPORAL FEATURE RESULTS

The resume analysis module presents the analysed profile through different indicators such as resume quality, ATS-oriented score, skill coverage and domain fit. The system also displays the skills detected from the resume and highlights areas that may require improvement. The analysis helps identify missing or weak sections that can be improved before applying for suitable job roles. It also provides a structured summary of the candidate's technical strengths based on the uploaded resume. The extracted information is further used by other modules for job matching and skill gap identification. This creates a clear starting point for the student to improve resume qu ality and career readiness. The analysis results provide a detailed view of how the resume represents the candidate's current profile. The ATS-oriented evaluation helps identify areas related to resume structure, relevant keywords and overall profile presentation. Skill coverage provides an indication of how many relevant technical

competencies are represented in the uploaded resume. Domain fit helps determine whether the identified skills are aligned with the selected technical area. The system also presents the detected skills in a structured manner so that the student can verify the information extracted from the resume.

Areas with lower coverage or weaker representation can be considered for further improvement. These results provide useful input for the subsequent job-description comparison process. The identified skills are reused when determining matching and missing skills for a selected role. The analysis therefore acts as the foundation for the skill-gap and recommendation modules. It also helps the student understand the current strengths of the resume before proceeding to further career assessment activities. Overall, the resume analysis module converts the uploaded document into meaningful profile-level information that supports the complete Career Compass AI workflow.

Analysis Component	Result / Output
Resume Quality	Overall quality indicator based on resume structure and content
ATS Score	ATS-oriented compatibility indicator
Skill Coverage	Percentage of relevant skills identified in the resume
Detected Skills	Technical and professional skills extracted from the resume
Domain Fit	Alignment with predefined technical domains
Resume Strengths	Strong areas identified from the resume
Improvement Areas	Sections or skills requiring further improvement

Table 6.1 – Resume Analysis Results

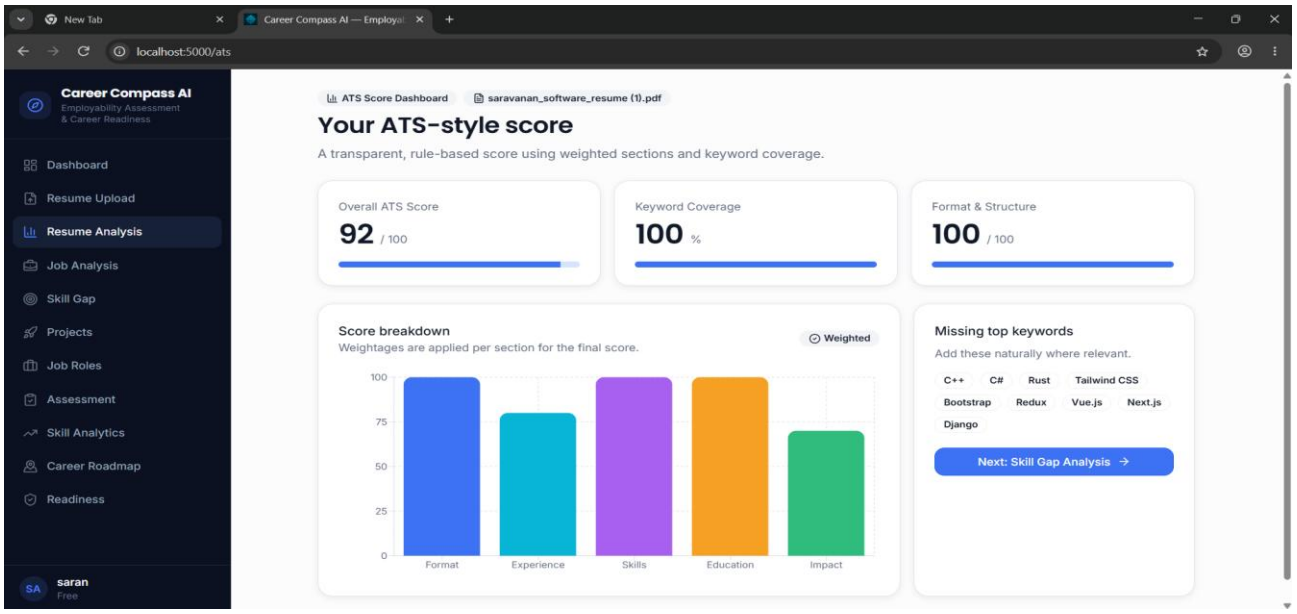


Figure 6.7 – Resume Analysis Dashboard

6.8 JOB DESCRIPTION ANALYSIS RESULTS

The job-description analysis module processes the selected job description and identifies important requirements such as technical skills, frameworks, databases, cloud technologies and other qualifications. This result acts as the reference profile for the subsequent resume–job matching process.The module helps identify the key competencies expected for the selected role. It also provides a structured representation of required and preferred skills from the job description. These extracted requirements are used to determine the skills already available in the student's profile. The analysis therefore supports accurate skill gap identification and role specific career recommen dations.

The analysed job requirements provide a structured reference for evaluating the student's suitability for the selected role. The extracted technical skills are normalized to support consistent comparison with the skills identified from the resume. This helps the system recognize common variations in technology names during the matching process. The module also separates important requirements so that essential competencies can receive appropriate priority during skill-gap analysis. The processed job information is passed to the resume–job matching module for identifying matching and missing skills. The identified missing requirements can then be used to generate relevant project recommendations and preparation activities.

The job-description results also provide the required context for generating role-specific assessment questions. Assessment performance can subsequently be connected with the identified requirements to determine areas that need further improvement. Thus, the job-description analysis module provides the employer-side requirements needed for the remaining stages of the Career Compass AI workflow. Overall, the module establishes a structured connection between the selected job role, candidate profile, skill-gap analysis and personalized career preparation.

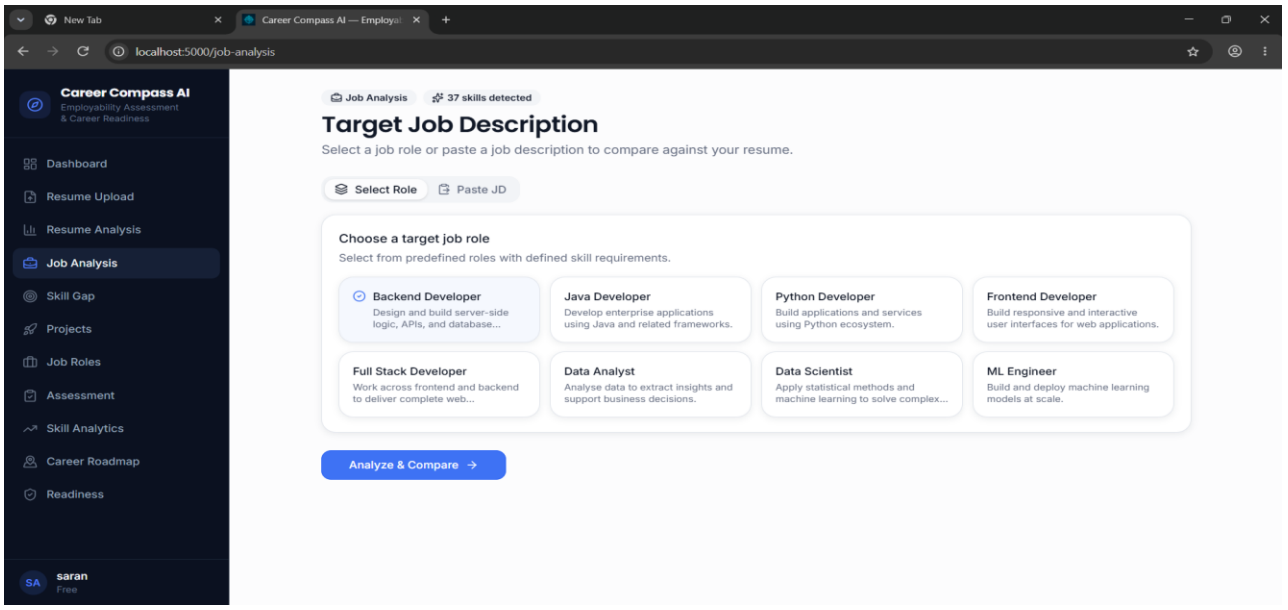


Figure 6.8 – Job Analysis Output

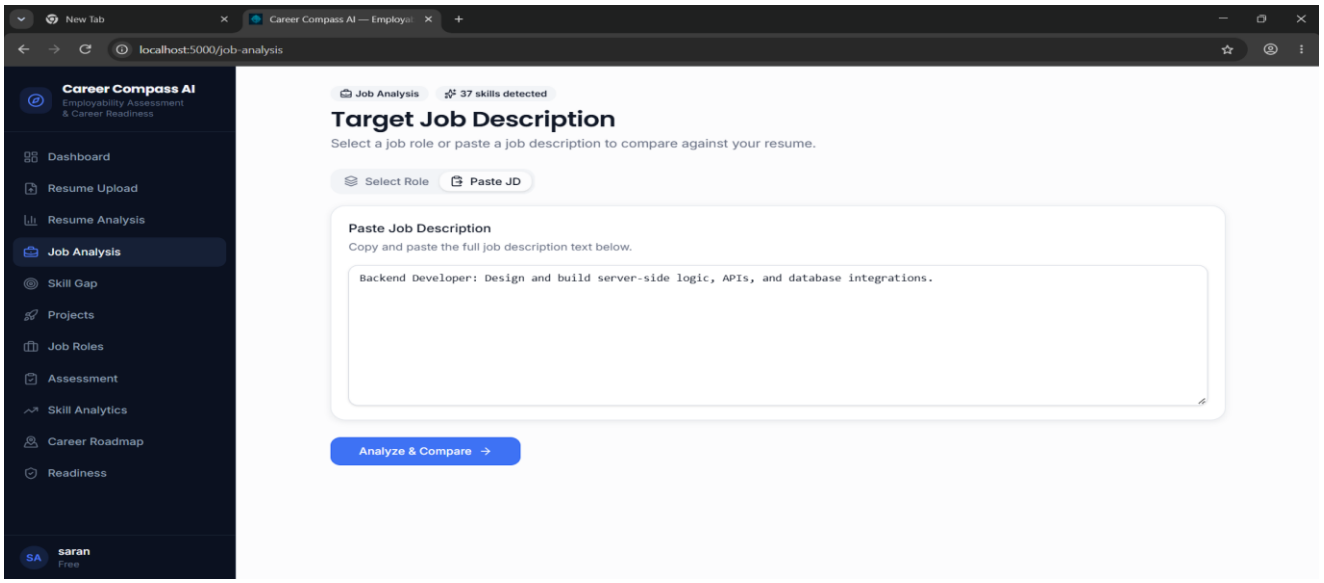


Figure 6.9 - Job Description Analysis Output

6.9 RESUME–JOB MATCHING RESULTS

The resume–job matching module compares the skills identified from the resume with the requirements extracted from the selected job description. Skills that occur in both profiles are presented as matching skills, while requirements that are not represented in the resume are identified as missing skills. The resulting comparison provides a more detailed view than a single overall match score because students can directly understand which technical areas are already covered and which areas need attention. The module considers the selected job role as the reference while performing the comparison. It helps identify the level of alignment between the candidate profile and the expected role requirements.

The matching results are presented in a structured manner for easier interpretation. Missing skills are highlighted so that students can focus on specific areas for improvement. The comparison also supports the generation of relevant project recommendations based on identified skill gaps. These results are further used by the assessment and career roadmap modules to provide role-specific guidance. Overall, the module converts the resume and job description into a clear skill-level comparison for career preparation.

The matching process uses the normalized skill information to improve consistency between the resume and job-description requirements. This allows equivalent or commonly used variations of technology names to be considered during comparison. The system provides a clear distinction between skills that are already available and those that require further development. This information helps the student understand the specific technical expectations of the selected role. The matching result can also be used to prioritize important missing competencies for further preparation.

The identified gaps are passed to the project recommendation module so that suitable practical projects can be suggested. Assessment performance can then provide additional evidence about the student's understanding of the identified skills. The matching output therefore serves as an important input for both practical skill development and knowledge assessment. It also supports the personalized roadmap by indicating which competencies should receive greater attention. Overall, the resume–job matching module connects candidate information with role requirements and converts the comparison into actionable career-preparation information.

Category	Description
Matching Skills	Skills available in both the resume and target job requirements
Missing Skills	Required skills not sufficiently represented in the resume
Improvement Skills	Skills that require additional development or stronger evidence
Skill Priority	Priority assigned according to the importance of the identified gap

Table 6.2 – Resume – Job Skill Matching Results

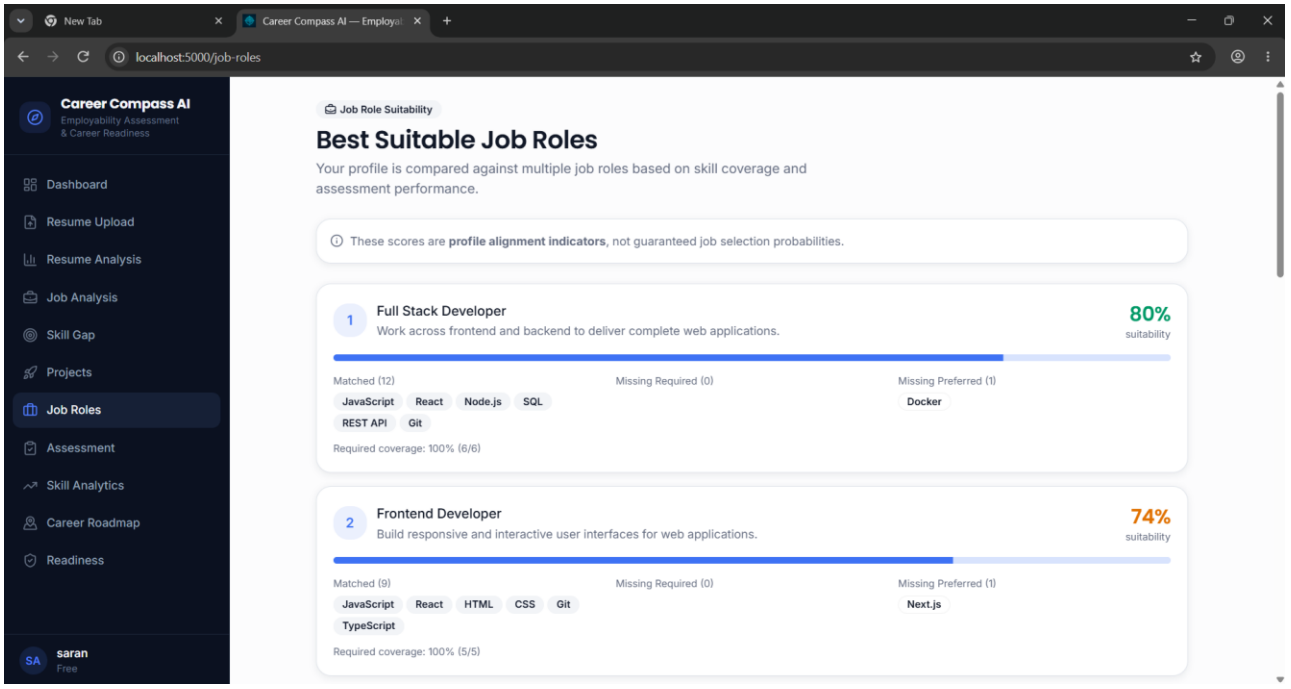


Figure 6.10 – Suitable Job Roles

6.10 SKILL GAP ANALYSIS RESULTS

The skill-gap module converts the matching result into a prioritized list of improvement areas. High-priority gaps represent important requirements that are not adequately covered, while medium- and low-priority gaps represent additional areas that can improve the student's profile. This result helps the student move from simply understanding the difference between the resume and job description to identifying which skills should be addressed first.

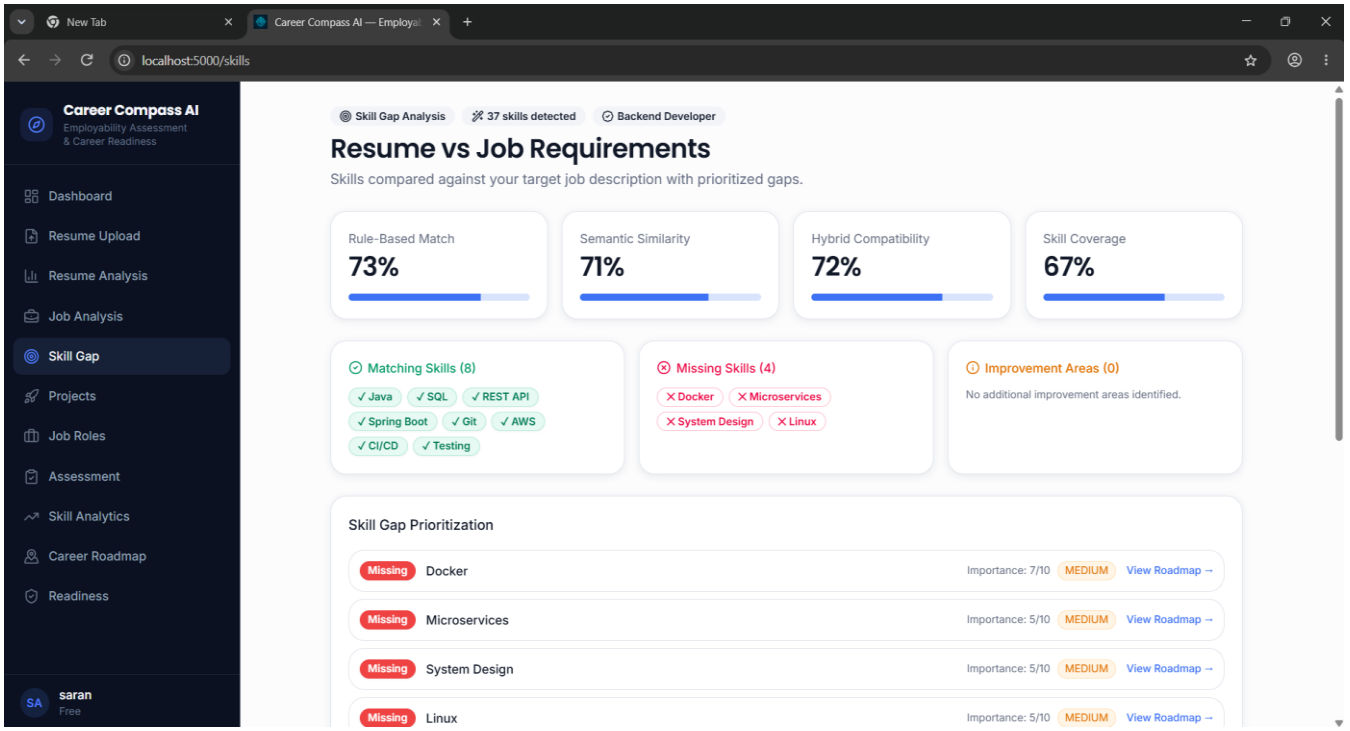


Figure 6.11 – Skill Gap Analysis

6.11 EVENT EVALUATION RESULT

The recommendation module uses the identified skill gaps to provide practical project suggestions and suitable job-role indicators. Project recommendations are associated with the skills they can help develop, allowing students to select projects that address multiple gaps. Job-role recommendations compare the student's current profile with predefined role requirements and present roles that have stronger alignment with the available skills. These recommendations are intended to support career preparation and should no be considered guaranteed employment predictions.

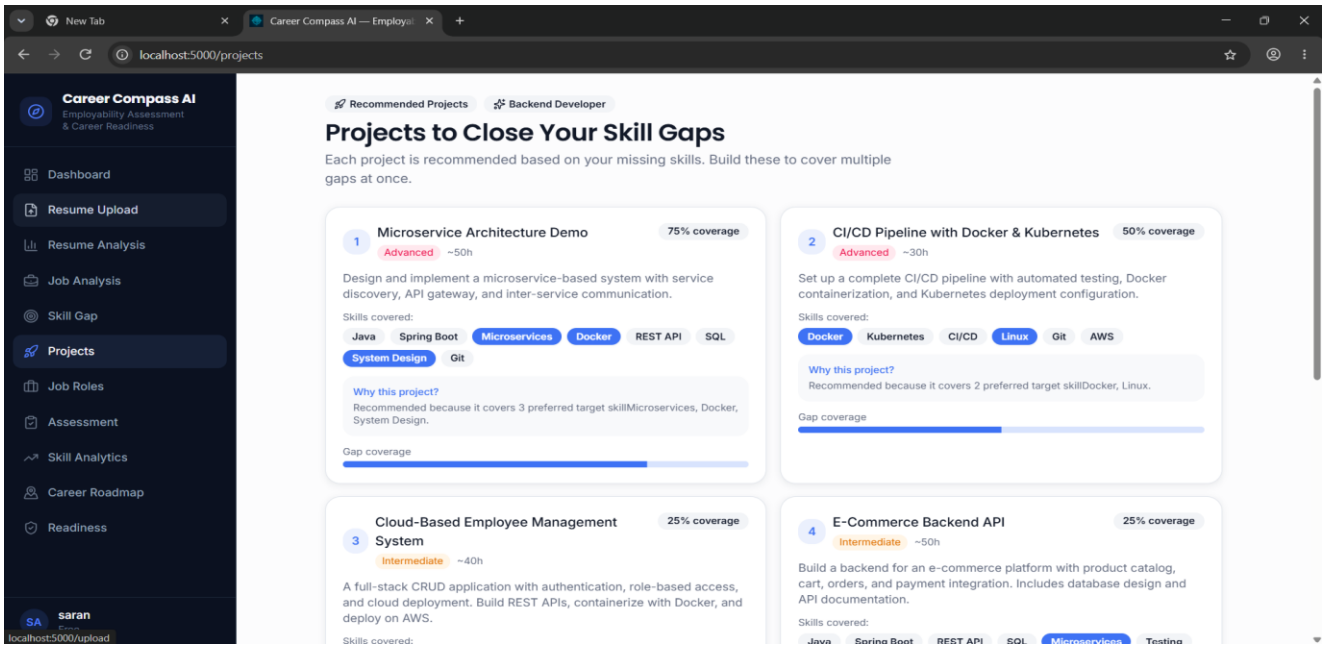


Figure 6.12 - Project Recommendation Output

6.12 MOCK ASSESSMENT RESULTS

The mock assessment module provides questions related to the selected job role and relevant technical skills. After the student submits the answers, the system calculates the assessment score and provides skill-wise performance information. The assessment result adds another level of evaluation because the system can compare the skills represented in the resume with the student's demonstrated performance during the assessment. The questions are designed to evaluate the student's understanding of the technical areas associated with the selected role. The assessment result provides a measurable indication of the student's current knowledge level. The system records the completed assessment and maintains the result for further analysis. Skill-wise performance helps identify areas where the student has performed well and areas that require additional preparation. The assessment score is also considered along with resume and matching results for overall career-readiness analysis. Previous assessment attempts can be used to observe changes in performance over time. This allows students to track their progress through repeated assessments. The results also support the generation of personalized learning recommendations. By combining profile information with assessment performance, the module provides a more complete view of the student's current readiness.

Parameter	Output
Total Questions	Number of questions presented in the assessment
Correct Answers	Number of correctly answered questions
Overall Score	Percentage-based assessment score
Skill-wise Score	Performance calculated for individual skills
Weak Skills	Skills with comparatively lower assessment performance
Strong Skills	Skills with comparatively higher assessment performance

Table 6.3 Mock Assessment Result

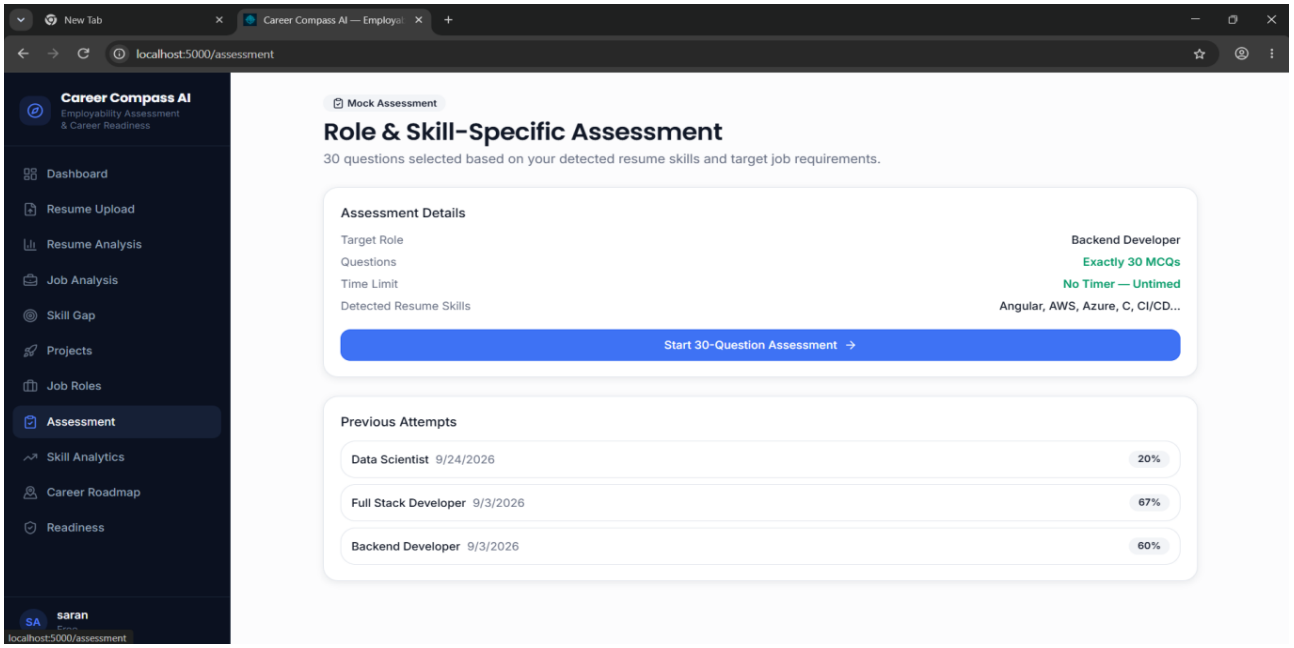


Figure 6.13 – Mock Assessment Interface

6.13 SKILL ANALYTICS AND ROADMAP RESULTS

The skill analytics module records assessment attempts and allows the student to compare performance over time. For example, if a student's performance in a particular skill increases between two assessment attempts, the improvement can be represented through charts and progress indicators. The roadmap module uses the identified gaps and assessment performance to prioritize learning activities. This creates a connection between assessment results and subsequent learning recommendations instead of providing a fixed learning path.

The identified gaps also provide a direct input for project recommendations and learning activities. Practical projects can be selected to help students develop multiple missing skills together. The skill-gap information is also used by the assessment module to provide questions related to the selected role and relevant technical areas. Assessment performance can further indicate whether the student has improved in the identified weak areas. The roadmap module uses these results to organize the required learning activities in a structured sequence.

This creates a connection between skill identification, practical development, assessment and continuous improvement. The module therefore transforms the resume–job comparison into specific and actionable preparation priorities. Overall, skill-gap analysis provides the foundation for personalized career development within Career Compass AI.

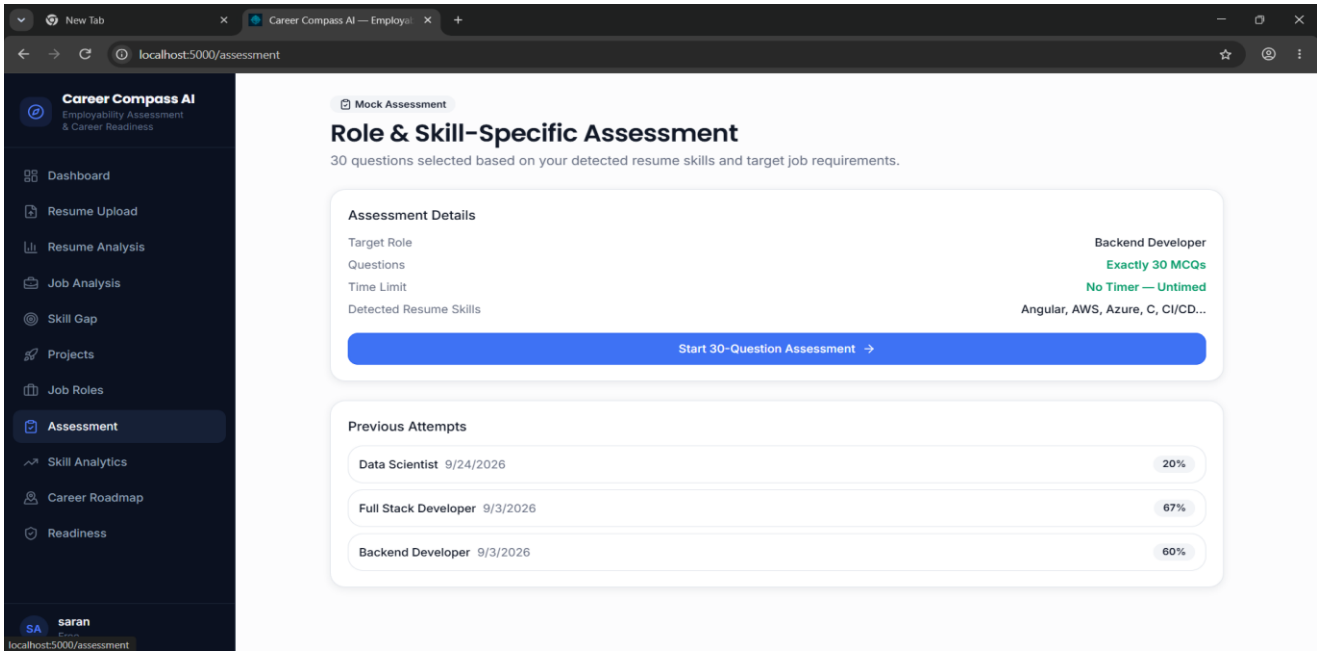


Figure 6.14 – Skill Progress Analytics

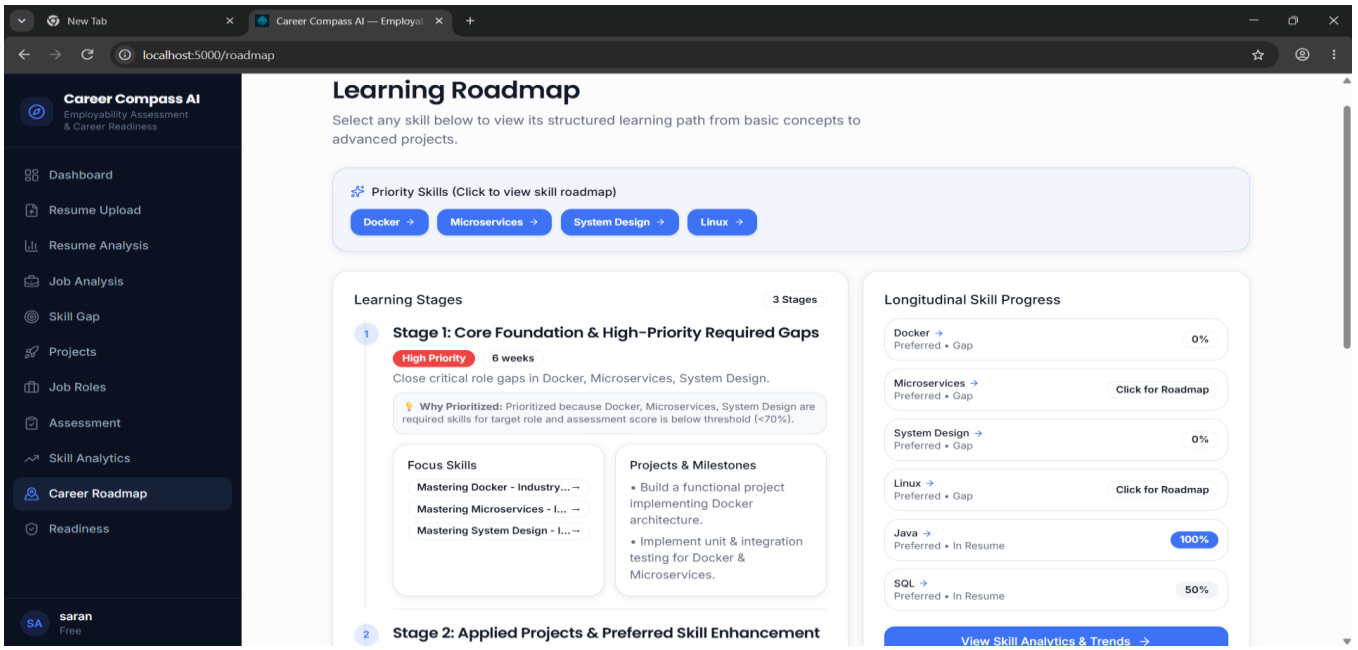


Figure 6.15 - Personalized RoadMap

6.14 READINESS INDICATOR RESULTS

The final stage presents a career-readiness indicator based on the defined profile alignment, skill coverage and assessment performance components. The indicator provides a consolidated view of the student's current preparation status for the selected role or profile. It is designed to help students understand their current position and identify remaining areas for improvement. The value represents a system-defined readiness or profile-alignment measure and is not intended to represent an actual probability of selection by a company. The result combines the outputs of the earlier analysis modules into a single reference indicator. It helps students monitor their progress and focus on areas that require further development. The indicator can also be reviewed after completing additional assessments or improving the resume profile.

The readiness indicator provides a simple way for students to review the overall outcome of the completed analysis process. It considers the relationship between the candidate profile and the requirements of the selected role. Skill coverage contributes information about the technical competencies represented in the student's profile, while assessment performance provides additional evidence of the student's current understanding.

The combined result helps students identify whether their preparation is progressing in the desired direction. The indicator can be used as a reference before beginning further learning activities and again after completing recommended improvements. Changes in the underlying resume, skill profile or assessment performance can therefore lead to changes in the readiness result. This allows the student to observe the effect of continuous preparation over multiple assessment cycles. The result also provides a concise summary of the information generated by the earlier modules.

It connects resume analysis, job matching, skill-gap identification and assessment into a common outcome. The readiness indicator therefore completes the Career Compass AI workflow by presenting the student's current preparation status in an understandable form. Overall, it supports continuous self-evaluation and helps students make informed decisions about the areas they need to improve before applying for suitable roles.

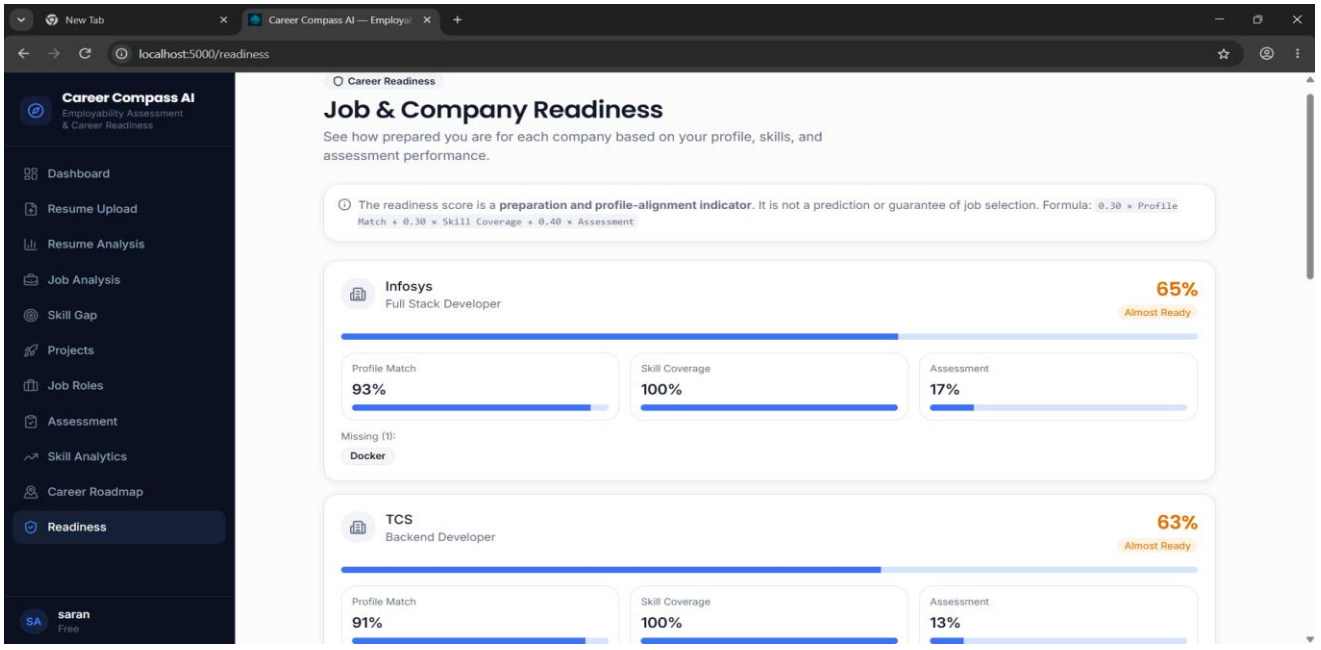


Figure 6.16 – Career Readiness Dashboard

CHAPTER 7

TESTING AND VALIDATION

7.1 TESTING METHODOLOGY

Testing is carried out to verify that the major modules of Career Compass AI work according to their intended functions and that the output generated by one module can be correctly used by the next module. The testing process covers authentication, resume processing, job-description analysis, skill matching, recommendations, assessment, analytics and roadmap generation. Functional testing is used to check individual features, integration testing verifies communication between connected modules, and validation checks whether the complete workflow produces meaningful outputs for the intended student use case.

7.2 UNIT TESTING

Unit testing focuses on individual functions and components of the system. Resume text processing, skill extraction, score calculation, skill matching, gap prioritization, project recommendation and assessment evaluation can be tested independently using suitable input cases. The purpose of unit testing is to identify errors within individual modules before they are combined with other components.

7.3 INTEGRATION TESTING

Integration Integration testing verifies whether the output of one module is correctly transferred to the next module. Similarly, identified skill gaps should be available to the project recommendation and roadmap modules, while assessment scores should be available to the analytics module. This testing ensures that the complete Career Compass AI workflow operates as an interconnected system.

7.4 API TESTING

API testing is used to verify backend endpoints responsible for authentication, resume processing, job-description management and other application operations. The tests check valid requests, invalid inputs, missing information and unauthorized access. The expected response structure, status codes and error handling are also verified to ensure reliable communication between the frontend and backend components.

7.5 FRONTED TESING

Frontend testing verifies whether the application pages, forms, navigation elements, upload components, charts and assessment interfaces work correctly. Different input conditions are checked to ensure that the interface provides appropriate feedback during successful operations and errors. The responsive layout and readability of important outputs such as skill gaps, assessment scores and roadmap information are also considered during validation

7.6 DATABASE AND SECURITY TESTING

Database testing verifies whether user-related information and application data are stored and retrieved correctly. Authentication testing checks valid login, invalid credentials, registration validation and protected operations. User ownership checks are important to ensure that one user's resume, job descriptions and assessment information cannot be accessed by another user. Input validation and secure password handling are also considered as part of the security testing process.

Phase	Test Area	Expected Result
1	User Authentication	Valid users can register and log in
2	Resume Processing	Resume text is extracted correctly
3	Resume Analysis	Relevant skills and analysis indicators are generated
4	Job Description Analysis	Job requirements and skills are identified
5	Resume–Job Matching	Matching and missing skills are displayed
6	Skill Gap Analysis	Skill gaps are assigned appropriate priorities
7	Recommendations	Relevant projects and job roles
8	Mock Assessment	Answers are evaluated

		and scores are generated
9	Skill Analytics	Assessment performance is tracked across attempts
10	Roadmap and Readiness	Prioritized roadmap and readiness indicator are generated

Table 7.1 Phase-Wise Testing Results

7.7 FINAL VALIDATION

Final validation is performed by executing the complete system workflow from user authentication through resume analysis, job-description analysis, skill-gap identification, recommendation, assessment, analytics and roadmap generation. The validation confirms whether the modules operate together and whether the information generated at an earlier stage is reflected in the subsequent stages. The final validation also checks the usability of the application and the consistency of the major outputs. Since the system is designed as a career-preparation platform, the final validation focuses on functional correctness and workflow continuity rather than treating the readiness indicator as a guaranteed prediction of employment.

Validation Area	Validation Objective
Authentication	Verify secure user access
Resume Processing	Verify document upload and text extraction
Resume Analysis	Verify profile and skill analysis
JD Analysis	Verify job requirement extraction
Skill Matching	Verify skill identification
Recommendations	Verify project and role recommendations
Assessment	Verify question evaluation and scoring

Analytics	Verify progress tracking
Roadmap	Verify prioritized learning guidance
Readiness	Verify consolidated preparation indicator

Table 7.2 – Final Validation Summary

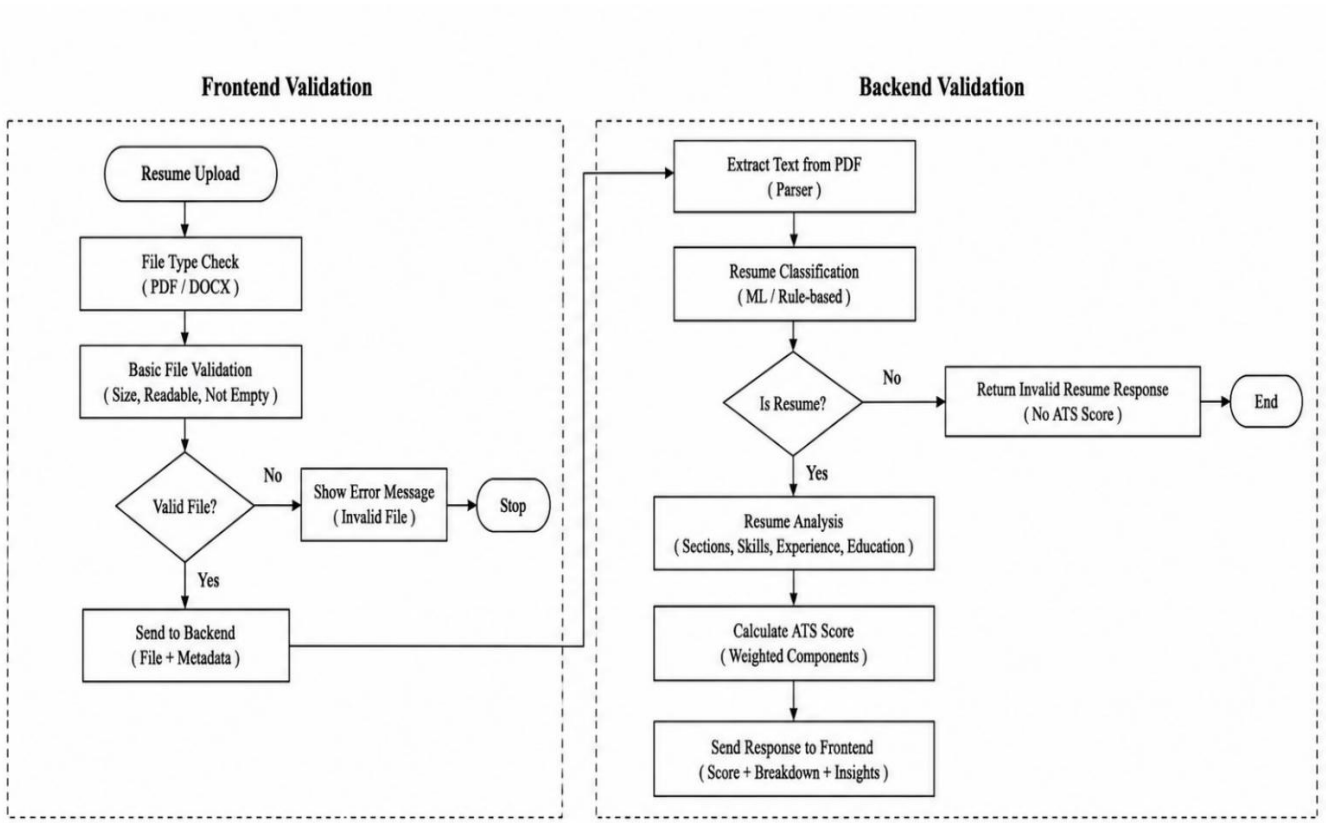


Figure 7.1 – Frontend And Backend Validation

Figure 7.1 – Frontend And Backend Validation

7.8 AUTHENTICATION AND AUTHORIZATION TESTING

Authentication and authorization testing was performed to verify that the Career Compass AI platform securely manages user access and prevents unauthorized users from accessing protected application features. The testing covered the complete authentication lifecycle, beginning with new user registration and continuing through login, authenticated session handling, protected route access, and logout. During registration testing, valid user details were entered and the system was checked to ensure that the account was created successfully and stored in the database. Login testing was then performed using valid and invalid credentials to verify that only authenticated users could establish an active session. The system was also tested with incorrect passwords and invalid login details to confirm that unsuccessful authentication attempts were handled properly without granting access. After successful login, protected modules such as Resume Upload, Resume Analysis, Job Analysis, Skill Gap Analysis, Projects, Assessment, Skill Analytics, Career Roadmap, and Readiness were accessed to verify that authenticated users could navigate through the complete application workflow. Authorization testing was additionally performed by attempting to access protected routes without an authenticated session. The system was expected to redirect unauthenticated users to the login interface rather than displaying protected application data. Session persistence was verified by navigating between different modules after login and confirming that the authenticated state remained available. Logout functionality was also tested to ensure that the active session was cleared correctly and that previously protected pages could no longer be accessed without logging in again. These tests verified the interaction between the frontend authentication state, backend authentication APIs, JWT-based authorization, and protected route handling. The results confirmed that authentication and authorization mechanisms were functioning as intended and that access to user-specific application functionality was controlled according to the user's authentication state.

7.9 DATABASE AND DATA PERSISTENCE TESTING

Database and data persistence testing was conducted to verify that user information and application-generated data are correctly stored, retrieved, updated, and maintained throughout the system lifecycle. The Career Compass AI platform uses SQLite with SQLAlchemy for persistent backend data management. Testing began by registering a new user and verifying that the corresponding account information was successfully stored in the database. After authentication, resume information, resume analysis results, job descriptions, job analysis results, assessment attempts, skill-related information, roadmap progress, and other user-specific records were checked to ensure

that the data generated during different stages of the application workflow was properly associated with the authenticated user. The system was also tested by uploading and analyzing a resume, selecting a job role or providing a job description, performing skill-gap analysis, and generating downstream recommendations. This helped verify that information generated in one module could be correctly retrieved and used by subsequent modules without unexpected data loss. Persistence testing was also performed by restarting the backend application and checking whether previously stored user and project-related information remained available after the restart. This ensured that the application was not relying only on temporary frontend state or in-memory values for important persistent information. User isolation was also considered during validation so that data belonging to one authenticated user would not incorrectly appear in another user's workflow. Database interactions were monitored for successful creation and retrieval of records, while invalid or missing identifiers were tested to ensure that the backend returned appropriate responses instead of producing application failures. These tests confirmed that the database layer was correctly integrated with the backend services and that important application data remained persistent across normal application restarts and user sessions.

7.10 ERROR HANDLING AND SECURITY VALIDATION

Error handling and security validation were performed to ensure that the application responds gracefully to invalid input, missing data, unauthorized requests, and unexpected user actions without causing page crashes or exposing internal system information. The testing covered different categories of invalid requests, including unauthenticated API requests, invalid authentication credentials, missing required fields, invalid identifiers, unsupported operations, and attempts to access protected functionality without an active session. Backend API responses were examined to verify that appropriate HTTP status codes and meaningful error messages were returned for different failure conditions The frontend was also tested to ensure that API failures were displayed through controlled messages or notifications rather than leaving the user with a blank page or broken interface. Security validation also included checking the authentication flow, protected routes, JWT-based session handling, and user-specific access control. The system was tested to ensure that sensitive application functionality was available only after successful authentication and that logout correctly removed the active authenticated state. Input validation was checked at relevant frontend and backend boundaries so that malformed data would not directly cause application crashes. The resume processing workflow was also validated to distinguish valid resume documents from unrelated documents before performing ATS

analysis. This validation prevents arbitrary documents such as academic materials or unrelated PDF files from being treated as resumes. Overall, the error-handling and security tests focused on maintaining application stability, providing understandable feedback to users, preventing unauthorized access, and ensuring that invalid inputs were handled in a controlled manner without affecting other modules.

7.11 END-TO-END SYSTEM VALIDATION

End-to-end validation was performed to verify the complete Career Compass AI workflow from user registration to the final career-readiness output. Instead of testing each module independently, the complete sequence was executed using realistic user interactions to confirm that data generated by one module was correctly transferred to the next module. The validation began with user registration and login, followed by resume upload and resume processing. The uploaded resume was parsed and analyzed to extract relevant candidate information, technical skills, education, experience, projects, and other available resume details. The ATS-oriented analysis was then generated from the validated resume content. The job analysis workflow was subsequently tested by selecting a predefined job role or providing a job description. The selected role's job requirements and detected skills were passed into the resume–job comparison process, where the candidate's extracted skills were compared with the target requirements using the implemented matching and semantic analysis process. The downstream workflow was then validated by checking the matching and missing skills produced by the skill-gap analysis and verifying that the identified missing skills were used for project recommendations. Assessment attempts were then submitted and the resulting scores were reflected in the Skill Analytics module. The roadmap workflow was checked to ensure that relevant skill-development stages were presented based on identified skill gaps and assessment performance. Finally, the Readiness module was accessed to verify that the accumulated resume, matching, and assessment information could be used in the final readiness analysis. Navigation between modules, API communication, database persistence, authentication state, error handling, and frontend rendering were observed throughout the complete workflow. The end-to-end validation confirmed that the major components of Career Compass AI were integrated into a continuous workflow and that the application could process information from the initial user input through resume analysis, job comparison, skill-gap identification, project recommendation, assessment, analytics, roadmap generation, and final readiness analysis without breaking the intended application flow.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

Career Compass AI provides an integrated approach for student employability assessment and career preparation by connecting resume analysis, job-description analysis, skill-gap identification, project recommendation, job-role recommendation, mock assessment, skill analytics and personalized roadmap generation. The system helps students understand their existing skills, compare them with the requirements of a target role and identify areas that require improvement. Instead of limiting the output to a resume or job-match score, the proposed workflow continues towards practical skill development and assessment. The inclusion of skill-wise progress tracking allows students to observe their improvement across repeated assessments, while the roadmap provides a structured direction for further preparation. Thus, the system establishes a continuous process of **Analyse → Identify → Practise → Assess → Improve → Reassess** and provides a consolidated view of the student's career-preparation status. The integrated workflow also connects the results generated by different modules so that each stage contributes to the next stage of preparation. Students can use the identified skill gaps to select suitable projects and learning areas. Assessment results further help them understand whether their preparation is improving over time. The system therefore supports a structured and continuous approach to employability preparation rather than providing only a one-time analysis.

8.2 FUTURE WORK

Future development can extend Career Compass AI with advanced NLP and large language model capabilities for deeper resume and job-description understanding. Semantic embeddings can be introduced to improve matching between equivalent or contextually related skills, while LLM-based assessment generation can create more varied questions based on the student's resume, target role and identified weaknesses. The roadmap can also be made more adaptive by automatically changing priorities after every assessment attempt. Additional features such as coding assessments, interview simulation, voice-based mock interviews, real-time labour-market skill analysis and personalized learning-resource recommendations can further extend the platform. Integration with cloud deployment and larger-scale databases can support multiple institutions and larger numbers of users. Future evaluation can also use real student datasets and defined performance metrics to measure the effectiveness

of skill recommendations, assessments and personalized roadmaps. The system can also be enhanced with multilingual resume processing to support students from different language backgrounds. Continuous integration and automated testing can improve the reliability of future releases. Institution-level dashboards can provide aggregated insights into student skill development and placement preparation. These extensions can make the platform more adaptive, scalable and useful for long-term career preparation. Future development can extend Career Compass AI with advanced NLP and large language model capabilities for deeper resume and job-description understanding. Semantic embeddings can improve matching between equivalent or contextually related skills, while LLM-based assessment generation can create varied questions based on the student's resume, target role and identified weaknesses. The roadmap can also become more adaptive by automatically changing learning priorities after each assessment attempt. Additional features such as coding assessments, interview simulation, voice-based mock interviews, real-time labour-market skill analysis and personalized learning-resource recommendations can further enhance the platform. The system can also be extended with multilingual resume processing to support students from different language backgrounds. Cloud deployment and scalable databases can enable the platform to support multiple institutions and larger numbers of users. Future evaluation can use real student datasets and defined performance metrics to measure the effectiveness of skill recommendations, assessments and personalized roadmaps. Institution-level dashboards and continuous progress monitoring can further support student development. These enhancements can make Career Compass AI more adaptive, scalable and useful for long-term career preparation.

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APPENDIX

SOURCE CODE

```
import React, { useState } from "react";
import { create } from "zustand";

export interface Skill {
  name: string;
  keywords: string[];
  category: string;
}

export interface ResumeAnalysis {
  score: number;
  skills: string[];
  strengths: string[];
  improvements: string[];
  keywordCoverage: number;
}

export interface JobAnalysis {
  title: string;
  requiredSkills: string[];
  preferredSkills: string[];
  technologies: string[];
}

export interface SkillMatchResult {
  matchedSkills: string[];
  missingSkills: string[];
  improvementSkills: string[];
  matchScore: number;
}
```

```
export interface SkillGap {
  skill: string;
  priority: "High" | "Medium" | "Low";
  importance: number;
}

export interface Project {
  title: string;
  description: string;
  skills: string[];
  difficulty: string;
}

export interface JobRole {
  title: string;
  requiredSkills: string[];
}

export interface Question {
  id: string;
  skill: string;
  question: string;
  options: string[];
  answer: string;
  difficulty: "Easy" | "Medium" | "Hard";
}


export function calculateRoleScore(
  userSkills: string[],
  requiredSkills: string[]
): number {

  if (
    requiredSkills.length === 0
  ) {
```

```
    return 0;
  }


  const recommendedRoles =
    recommendRoles(
      roles,
      resumeAnalysis.skills
    );

  const roadmap =
    createRoadmap(
      skillGaps
    );

  const readiness =
    calculateReadiness(
      matching.matchScore,
      resumeAnalysis.keywordCoverage,
      0
    );

  return {
    resumeAnalysis,
    jobAnalysis,
    matching,
    skillGaps,
    recommendedProjects,
    recommendedRoles,
    roadmap,
    readiness
  };
}

export function saveAssessmentAttempt(
  history: AssessmentAttempt[],
```

```
  attempt: AssessmentAttempt
) {

  return [
    ...history,
    attempt
  ];
}

export function getLatestAssessment(
  history: AssessmentAttempt[]
) {

  if (
    history.length === 0
  ) {
    return null;
  }

  return history[
    history.length - 1
  ];
}

export function compareAssessments(
  previous: AssessmentAttempt,
  latest: AssessmentAttempt
) {

  return {
    previousScore:
      previous.score,

    latestScore:
      latest.score,
```

```
    improvement:
      latest.score -
      previous.score
  };
}

export async function processResume(
  file: File
) {

  if (
    file.type !==
    "application/pdf"
  ) {

    throw new Error(
      "Please upload a PDF resume."
    );
  }

  const buffer =
    await file.arrayBuffer();

  return buffer;
}

export function validateJobDescription(
  description: string
) {

  if (
    !description ||
    description.trim().length < 20
  ) {

    return {
```

```
      valid: false,
      message:
        "Please provide a valid job description."
    };
  }

  return {
    valid: true,
    message:
      "Job description is valid."
  };
}

export function generateImprovementMessage(
  skill: string,
  score: number
) {

  if (score < 40) {

    return (
      `Your current performance in ${skill} `
      + `is low. Start with the fundamentals `
      + `and practise using small exercises.`
    );
  }

  if (score < 70) {

    return (
      `You have basic knowledge of ${skill}. `
      + `More practical practice is recommended.`
    );
  }

  return (
```

```
    `Your performance in ${skill} `
    + `is comparatively strong. Continue `
    + `with advanced practice and projects.`
  );
}

export function generateRoadmapSummary(
  gaps: SkillGap[]
) {

  if (
    gaps.length === 0
  ) {

    return (
      "No major skill gaps were identified. "
      + "Continue practising and reassess your profile."
    );
  }

  const highPriority =
    gaps.filter(
      gap =>
        gap.priority ===
        "High"
    );

  return (
    `Focus first on ${highPriority.length} `
    + `high-priority skill areas, followed by `
    + `medium and low-priority improvements.`
  );
}

export function calculateSkillCoverage(
  matchedSkills: string[],
```

```
  totalRequiredSkills: number
) {

  if (
    totalRequiredSkills === 0
  ) {
    return 0;
  }

  return Math.round(
    (matchedSkills.length /
      totalRequiredSkills) * 100
  );
}

export function calculateAssessmentScore(
  correctAnswers: number,
  totalQuestions: number
) {

  if (
    totalQuestions === 0
  ) {
    return 0;
  }

  return Math.round(
    (correctAnswers /
      totalQuestions) * 100
  );
}

export function classifyPerformance(
  score: number
) {
```

```
  if (score >= 80) {
    return "Strong";
  }

  if (score >= 60) {
    return "Moderate";
  }

  return "Needs Improvement";
}

export function getPriorityMessage(
  priority:
    "High" | "Medium" | "Low"
) {

  switch (priority) {

    case "High":
      return (
        "Important skill. "
        + "Prioritize this area."
      );

    case "Medium":
      return (
        "Relevant skill. "
        + "Plan regular practice."
      );

    default:
      return (
        "Additional skill. "
        + "Can be improved later."
      );
  }
```

```
}

export function createAssessmentReport(
  questions: Question[],
  answers: Record<string, string>
) {

  const result =
    evaluateAssessment(
      questions,
      answers
    );

  return {
    ...result,

    performance:
      classifyPerformance(
        result.score
      ),

    generatedAt:
      new Date().toISOString()
  };
}

export function createCareerReport(
  resumeText: string,
  jobDescription: string
) {

  const workflow =
    runCareerCompassWorkflow(
      resumeText,
      jobDescription
    );
```

```
  return {
    resumeScore:
      workflow.resumeAnalysis.score,

    jobMatch:
      workflow.matching.matchScore,

    matchingSkills:
      workflow.matching
        .matchedSkills,

    missingSkills:
      workflow.matching
        .missingSkills,

    skillGaps:
      workflow.skillGaps,

    projects:
      workflow.recommendedProjects,

    roles:
      workflow.recommendedRoles,

    roadmap:
      workflow.roadmap,

    readiness:
      workflow.readiness
  };
}
```

Career Compass AI: An AI-Powered Employability Assessment and Career Readiness Platform Using Resume Analysis and Skill Gap Assessment

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***Abstract*—**
Students often learn several technologies during their studies, but they may not know how closely their current skills match the requirements of a particular job. A resume can contain programming languages, frameworks, projects, internships, and certifications, but these details alone do not show whether the student has enough practical knowledge. Career Compass AI is proposed as an integrated platform that connects resume analysis, job matching, skill-gap identification, project recommendation, technical assessment, progress tracking, and career-roadmap generation.

The system accepts a student's resume together with a job description or selected target role. Important information is extracted and organized before the profile is compared with the selected requirements. The analysis then identifies matched skills, missing skills, and areas that need improvement. Important skill gaps are connected with practical project suggestions so that students receive a clearer direction for developing the required competencies.

The platform also provides role-specific assessments to obtain additional evidence of technical understanding. Previous assessment attempts are retained so that changes in performance can be observed over time. Based on the latest results, learning priorities and the career roadmap can be updated. In this way, Career Compass AI treats employability preparation as a continuous process of analysis, learning, assessment, and reassessment rather than as a one-time resume evaluation.

***Index Terms*—Career readiness, resume analysis, skill gap analysis, job matching, natural language processing, technical assessment, project recommendation, career roadmap**

I. Introduction

During their studies, students tend to gain knowledge of several technologies without understanding the correlation of their skills to the job requirements. A resume may include details such as programming languages, frameworks, projects, internships, and certificates, but this information alone does not guarantee practical knowledge. Career Compass AI is a comprehensive system offering services such as resume analysis, job matching, skills gap identification, project suggestions, technical assessments, progress tracking, and career roads planning.

The system accepts a student resume and the job description of a target position. Then, relevant information is extracted from the resume and organized before the resume is compared with the selected job requirements. The analysis identifies the skills matching job requirements, lacking skills, and the skills requiring improvement. Along with necessary skills gap information, students get appropriate project recommendations.

The platform provides assessments based on the specific role to ensure more evidence of technical knowledge. The history of assessments is preserved so that progress can be monitored. Following new results, learning priorities and the career pathway can be changed. Thus, Career Compass AI treats employability preparation as a perpetual process of analysis and learning rather than a singular examination of one's resume.

Preparing for campus placements can be challenging because students need to do more than just create a resume. They have to understand the required skills for a job, pinpoint their shortcomings in their profile, and choose what should be learned prior to applying for the position. This process becomes even more complicated when students look for options among various job titles because different positions might require different sets of programming languages, frameworks, databases, cloud technologies, and methodologies.

A resume contains relevant information about education and qualifications in terms of skills, projects worked on, internships, and certificates. However, listing a technology does not mean having enough experience to work with it. The wording used in a job description may differ from similar requirements. For instance, one company may use the term "RESTful services" while another one will refer to "REST API". An algorithm based only on exact wording may miss the necessary correlations between different skills.

Techniques provided by Natural Language Processing play a significant role in the processing this form of unstructured information. The implementation of Transformer-based frame-

works led to the introduction of contextual representations that could help resolve many issues facing people that require the processing of information in the field of natural language processing (NLP) [3]. BERT demonstrated the power of bidirectional into contextual representations for the improvement of understanding language [4]. Sentence-BERT made it possible to compare the meaning of sentences and fragments of texts in the least complicated manner, with the help of meaningful sentence embeddings [5]. The use of information retrieval technologies allows to represent documents and calculate their value [1].

The system links profile analysis with competence development. It matches the learner's profile with the selected goal position, pinpointing important qualifications gaps, proposing relevant projects to fill them and assessing the knowledge gained through specialized evaluations. Past results are saved for tracking progress and modifying future priorities.

II. Related Work

Automated recruitment as well as career services usually employ information retrieval, natural language processing, semantic similarity, machine learning as well as recommendation techniques. Information retrieval has provided the well-known techniques to describe documents and evaluate their relevancy to queries or documents related [1]. The learning to rank approaches also provide a good example of how to learn relevancy decisions.

It might also be true that transformers advanced the real capacity of NLP systems concerning the contextual relevance of words. BERT optimized this concept by allowing the use of bidirectional pre-training and getting impressive results in several cases of language comprehension. The Sentence-BERT method emerged later and allowed for comparing sentences using semantic embeddings. The rationale behind these methods is useful for employability application, mostly because the skills and responsibilities of a given position might be expressed by using different wording.

Random Forest has an ensemble learning method aimed at classifications and predictions [6]. XGBoost allows for creating an effective gradient boosting model for structured data [7]. This technique may be considered for willingness assessment when a labelled dataset is present.

Generally, existing methods relate only to different levels of career preparation process. For instance, a resume parsing software does not instruct the user on how to enhance their profile; a job matching system offers merely a score of similarity, while the way to proceed after that remains unclear; an evaluation tool assesses technical knowledge without linking it to the targeted job of a student.

In contrast, Career Compass AI integrates these steps into one workflow that allows conducting resume analysis, job matching, skill gap identification, project implementation, technical assessment, tracking progress, and adapting a roadmap.

III. Research Gap

The primary issue that this paper addresses is the gap in linking the resume-job match analysis with actual skill development.

Profile Analysis → Skill Diagnosis → Skill Development → Evaluation → Reevaluation

With this system the preparation stage of the student can be updated as new project or assessment evidence appears.

IV. Proposed System

Career Compass AI operates as a modular system with the result obtained from one step being used as an input for another step. The process starts from a student's CV and a job description or position of interest.

Currently available features are similar to a preparatory system rather than a one-time scoring system. Upon completion of the initial stage of analysis the produced outcomes are utilized to provide practical recommendations. Assessment results enter the process later on and are used to update the recommendations.

The main stages of Career Compass AI consist of resume analysis, job description analysis, normalization of skills, resume-job matching, identification of skill gaps, suggestion of projects, suggestion of a position, technical assessment, progress tracking, and building of a roadmap.

In the first phase, the resume of a student is converted into information that can be used by the remaining modules.

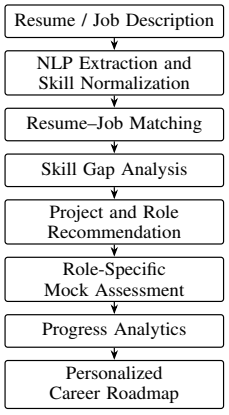


Fig. 1. Processing workflow of the proposed Career Compass AI framework.

V. Methodology

A. Resume Processing

The first stage converts the student's resume into information that can be used by the remaining modules. Resumes normally contain both structured headings and free-form descriptions. The system therefore extracts text and identifies sections such as education, technical skills, projects, internships, certifications, and achievements.

For machine-readable documents, text can be extracted directly. When scanned pages are present, OCR can be applied before further processing. The extracted text is cleaned by removing unnecessary spaces, repeated formatting characters, and other artifacts.

The resulting profile contains role-relevant attributes such as programming languages, tools, frameworks, databases, project experience, certifications, and internship information.

B. Job Description Processing

The job-description module follows a similar process. It identifies programming languages, frameworks, databases, cloud services, tools, and other competencies mentioned in the selected requirement.

Where sufficient information is available, requirements can be separated into mandatory and preferred categories. This helps the system prioritize important requirements over optional ones.

C. Skill Normalization

Skill normalization is important because the same competency may appear in several forms. Without normalization, a system may incorrectly identify a skill as missing even when an equivalent expression is present.

For example:

$$REST\ API,\ REST ful\ API,\ REST\ Services \rightarrow REST\ API \tag{1}$$

Similarly:

$$Amazon\ Web\ Services,\ AWS \rightarrow AWS \tag{2}$$

A normalization dictionary can map abbreviations, spelling variations, and common alternative expressions to a standard skill identifier.

D. Resume–Job Matching

Let S_r represent the normalized skills obtained from the resume and S_j represent the requirements obtained from the job description. A basic skill coverage value can be calculated as:

$$C = \frac{|S_r \cap S_j|}{|S_j|} \times 100 \tag{3}$$

where C represents the percentage of identified job requirements covered by the profile.

Keyword overlap may not be sufficient when different wording is used. Semantic representations can therefore be used in addition to direct matching. Cosine similarity between two embeddings is represented as:

$$Sim(E_r, E_j) = \frac{E_r \cdot E_j}{\|E_r\|\|E_j\|} \tag{4}$$

Here, E_r and E_j represent the corresponding vector representations.

VI. Skill Gap Analysis

After comparing the resume with the target role, the system identifies competencies that require attention. Instead of treating every unmatched term equally, the proposed approach divides the results into matched, missing, and improvement categories.

A matched skill is a competency represented in the student's profile. A missing skill is a requirement for which sufficient evidence is not found. An improvement skill is already present but may require stronger practical evidence or deeper understanding.

The priority of a gap depends on its importance to the selected role and the size of the observed gap. A conceptual priority score is:

$$P_s = R_s \times G_s \times I_s \tag{5}$$

where R_s represents role importance, G_s represents the estimated gap, and I_s represents relevance to the target position.

TABLE I
ILLUSTRATIVE SKILL GAP ANALYSIS

Skill	Status	Priority
Java	Match	–
Spring Boot	Match	–
MySQL	Match	–
Docker	Missing	High
AWS	Missing	High
Spring Security	Improve	Medium
Git	Improve	Medium

Note: The values shown are illustrative examples and are not experimental results.

VII. Project Recommendation

A conventional skill-gap report can identify what a student is missing but may not explain how that competency can be developed. Career Compass AI addresses this limitation by connecting identified gaps with practical project templates.

For example, when Docker, AWS, Spring Boot, and REST API development are identified as important gaps for a backend-oriented role, the platform can recommend an Employee Management System. The project can include a Spring Boot backend, MySQL database, REST endpoints, Docker containerization, Git-based version control, and cloud deployment.

The recommendation process can therefore be represented as:

Missing Skill → Project Template → Implementation Task → Skill Evidence

The purpose is not simply to increase the number of projects in a resume. The recommended project should provide an opportunity to apply the missing technology and create practical evidence of learning.

TABLE II
EXAMPLE PROJECT-TO-SKILL MAPPING

Project Component	Skill Developed
Backend Development	Spring Boot
Database Layer	MySQL
API Layer	REST API
Security	Spring Security
Containerization	Docker
Cloud Deployment	AWS
Version Control	Git

VIII. Suitable Job Role Recommendation

Students do not always have a single target position. A profile that is suitable for one role may require additional preparation for another. Career Compass AI can therefore compare the same profile with multiple predefined job roles.

The role recommendation output is intended to indicate preparation alignment rather than predict whether a company will hire a student. The comparison considers matched skills, missing requirements, and, where available, assessment performance.

TABLE III
ILLUSTRATIVE JOB ROLE SUITABILITY

Job Role	Match	Level
Backend Developer	92%	Strong
Java Developer	88%	Strong
Full Stack Developer	81%	Good
Data Analyst	68%	Moderate
AI Engineer	60%	Improve

Note: These values are illustrative and should not be reported as measured experimental performance.

IX. Role-Specific Mock Assessment

A resume can indicate that a student has used a particular technology, but it cannot independently verify the level of understanding. The proposed assessment module provides an additional source of evidence through role-specific questions.

The assessment is selected according to the target role and important skills identified during the earlier analysis. For a Backend Developer profile, the question set may include Java, Spring Boot, SQL, REST APIs, Docker, Git, and cloud fundamentals.

The assessment process identifies the target role, obtains high-priority skills, selects suitable questions, conducts the test, calculates overall and skill-wise scores, and stores the attempt for later comparison.

This distinction between a skill being mentioned and a skill being demonstrated is an important part of the proposed framework.

TABLE IV
ILLUSTRATIVE SKILL-WISE ASSESSMENT HISTORY

Skill	Test 1	Test 2	Change
Java	65%	88%	+23
Spring Boot	40%	75%	+35
SQL	78%	84%	+6
Docker	20%	70%	+50
AWS	30%	65%	+35

X. Skill Analytics and Progress Tracking

Career preparation becomes more meaningful when changes can be observed over time. The proposed system stores individual assessment attempts instead of retaining only the latest score.

For a particular skill s, the change between two attempts can be represented as:

$$\Delta_s = S_{latest,s} - S_{previous,s} \tag{6}$$

where $S_{latest,s}$ is the latest score and $S_{previous,s}$ is an earlier score.

For example, an increase in the Docker assessment score from 20% to 70% corresponds to a 50 percentage-point improvement in assessment performance. This should not be interpreted as a direct measurement of real-world proficiency.

The analytics component can display the initial score, latest score, change over time, strongest skill, weakest skill, and remaining high-priority gaps. This gives students a simple way to decide whether their current preparation strategy needs to continue or change.

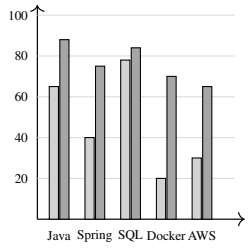


Fig. 2. Illustrative comparison of skill scores across assessment attempts.

XI. Personalized and Adaptive Career Roadmap

The roadmap is the stage where the analysis becomes a practical preparation plan. Instead of presenting students with a long list of unrelated learning resources, the system organizes activities around skills that are important for the selected role.

A typical roadmap begins with understanding the fundamentals of an identified skill, followed by suitable learning material and a practical task. The student can then apply the skill in a project, complete a role-specific assessment, review the result, and reassess the remaining gaps.

For example, if a student has reasonable Java knowledge but performs poorly in Docker and AWS, the next roadmap stage can focus on Docker fundamentals, containerizing a Spring Boot application, understanding basic AWS deployment concepts, and repeating the assessment.

The roadmap therefore changes according to newly observed evidence rather than remaining fixed after the initial resume analysis.

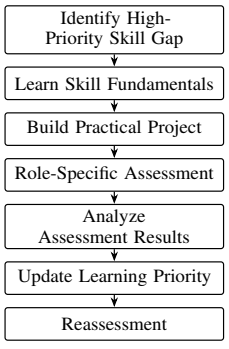


Fig. 3. Adaptive career-roadmap sequence driven by skill-gap and assessment feedback.

XII. Preparation Readiness Model

The system can provide a preparation-readiness indicator by combining observations from the student's profile, target role, and assessment performance. Unlike a simple resume

similarity value, the indicator considers multiple sources of evidence.

A conceptual formulation is:

$$R = w_1 M + w_2 C + w_3 A \tag{7}$$

where M represents resume-job matching, C represents important skill coverage, A represents assessment performance, and w_1, w_2, w_3 are configurable weights.

The weights satisfy:

$$w_1 + w_2 + w_3 = 1 \tag{8}$$

The indicator is intended to describe preparation status rather than predict employment. Before being treated as a calibrated measure, the formulation should be tested with a suitable dataset.

Recruitment decisions depend on several factors outside the scope of this model, including interviews, coding performance, communication ability, work experience, and organization-specific selection procedures.

XIII. System Architecture and Implementation

The proposed platform follows a modular web architecture. The user interface provides facilities for resume upload, target-role selection, analysis results, assessment interaction, progress visualization, and roadmap presentation.

The application layer coordinates communication between the interface and processing modules. NLP processing handles resume and job-description text, while the matching engine compares normalized skills and semantic representations. The skill-gap module identifies missing and improvement areas. The recommendation engine connects these gaps with suitable projects, while the assessment engine manages role-specific tests.

A persistent database retains user profiles, extracted skills, target roles, assessment attempts, recommendations, and roadmap information. This structure allows previous results to be used when generating later recommendations.

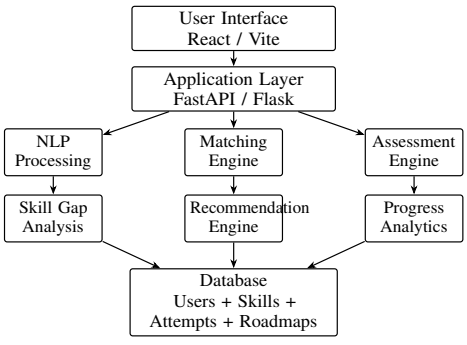


Fig. 4. Modular architecture of the proposed Career Compass AI platform.

A. Technology Stack

The implementation can use React with Vite for the web interface and a Python-based backend such as FastAPI or Flask for application services. Python libraries can support document processing, NLP operations, semantic comparison, and machine-learning components.

A relational database such as MySQL or PostgreSQL can store user profiles, extracted skills, job requirements, assessment attempts, project recommendations, and roadmap information. Chart.js or Recharts can be used to display skill-progress information through simple visualizations.

TABLE V
PROPOSED TECHNOLOGY STACK

Layer	Technology
Frontend	React.js, Vite
UI Styling	CSS / Tailwind CSS
Backend	FastAPI / Flask
NLP	Python, NLP Libraries
ML	Scikit-learn
Database	MySQL / PostgreSQL
Visualization	Chart.js / Recharts
Version Control	Git / GitHub

XIV. Data Model and Processing Flow

The database is designed to preserve information required for repeated analysis. A student may upload more than one resume and evaluate the profile against different target roles. Each target role can contain several required skills, and each assessment can generate multiple skill-wise results.

The principal entities include the user, resume, job profile, skill, skill gap, assessment, attempt, recommendation, and roadmap. The user entity stores basic account information, while the resume entity stores uploaded resume information and extracted content. The job profile contains the selected role and job-description information.

Skill-gap records maintain matched, missing, and improvement classifications. Assessment and attempt records preserve test information and individual results. Recommendation and roadmap records store suggested projects and learning priorities.

Maintaining these relationships allows the system to use earlier information when generating later recommendations. Assessment history, in particular, provides evidence for determining whether a skill area is improving or continues to require attention.

XV. Evaluation Methodology

Evaluation of Career Compass AI should be performed at both module and end-to-end levels. Since the platform performs several different tasks, a single accuracy value would not adequately describe its overall behaviour.

A. Resume Extraction Evaluation

Resume extraction can be evaluated using manually annotated resumes. Precision, recall, and F1-score can be calculated for information such as technical skills, education, experience, and certifications.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall} \tag{11}$$

These measures indicate how closely automatically extracted information agrees with manually verified information.

B. Skill Matching Evaluation

For skill matching, resume-job pairs can be manually annotated. Each requirement can be labelled according to whether the student's profile satisfies it. System predictions can then be compared with these reference labels.

C. Recommendation Evaluation

Project recommendations should be evaluated according to relevance. Evaluators can examine whether a recommended project addresses the identified skill gap and provides a realistic opportunity to apply the missing competency.

D. Assessment Evaluation

The assessment module can be evaluated in terms of question relevance, skill coverage, difficulty consistency, and repeated-attempt behaviour. Questions should remain aligned with the competencies expected for the selected role.

TABLE VI
EVALUATION PARAMETERS AND THEIR PURPOSE

Parameter	Evaluation Purpose
Resume Extraction	Correct extraction of profile entities
Skill Identification	Correct detection of relevant skills
Job Matching	Alignment with job requirements
Gap Detection	Identification of missing competencies
Recommendation	Relevance of suggested projects
Assessment	Reliability across attempts
Roadmap	Alignment with identified weaknesses
Progress Tracking	Measurement of score changes

XVI. Baseline and Ablation Study

An ablation study can be used to understand the contribution of the major modules. A simple baseline can begin with resume-job matching alone. Additional components can then be introduced progressively.

The first configuration performs profile matching only. The second adds skill-gap identification. The third includes role-specific assessment, while the final configuration adds the adaptive roadmap. The purpose is to determine whether the additional modules provide useful information beyond a conventional resume-matching approach.

TABLE VII
ABLATION CONFIGURATIONS

ID	Match	Gap	Test	Roadmap	Purpose
A	Yes	No	No	No	Baseline
B	Yes	Yes	No	No	Gap
C	Yes	Yes	Yes	No	Assessment
D	Yes	Yes	Yes	Yes	Complete

XVII. Results and Analysis Plan

The numerical performance of the proposed system should be reported only after testing with an appropriate dataset. The illustrative values shown in this paper demonstrate the intended presentation and should not be interpreted as measured experimental results.

The planned evaluation can include resume-extraction precision, recall and F1-score, skill-identification accuracy, agreement between automated job matching and human annotation, relevance of identified skill gaps, usefulness of project recommendations, assessment consistency, and changes in skill-wise scores across repeated attempts.

An important experiment is to compare the roadmap before and after a student completes an assessment. If the assessment shows improvement in one skill while another remains weak, the next roadmap should reflect that change. This provides a direct way to examine whether the adaptive component responds to newly available evidence.

XVIII. Discussion

The main strength of Career Compass AI is that it connects several stages of career preparation rather than treating resume analysis as the final result. A matching score can indicate how closely a profile resembles a job requirement, but it does not explain how the student can improve the profile.

The proposed workflow addresses this issue by converting identified gaps into practical development activities. When an important skill is missing, the system can prioritize it and connect it with a suitable project. The student can then use the project to gain implementation experience before taking another assessment.

The assessment component adds another useful distinction. A technology appearing in a resume is not necessarily equivalent to demonstrated knowledge. By evaluating role-specific questions and retaining previous attempts, the system obtains an additional signal that can be used when generating the roadmap.

The framework can also support multiple career directions. A student may have stronger preparation alignment with a Backend Developer role than an AI Engineer role. Instead of forcing a single recommendation, the system can show relative preparation levels and identify skills required to move toward another role.

The proposed framework should therefore be viewed as a career-preparation support system rather than an automated recruitment decision system.

XIX. Limitations

The proposed framework has several limitations. The first is the quality of input documents. Poorly formatted resumes, scanned pages, unusual layouts, or ambiguous terminology can reduce the accuracy of text and skill extraction.

A second limitation is that semantic similarity cannot prove practical proficiency. A student may mention a technology in a resume without having sufficient hands-on experience. The assessment module provides an additional signal, but even a structured test cannot completely replace practical project evaluation.

Recommendation quality also depends on the underlying skill-to-project knowledge base. If a new technology or specialized role is not adequately represented, the platform may produce a less useful recommendation.

The preparation-readiness indicator should not be interpreted as a probability of employment. Actual recruitment depends on interviews, coding ability, communication, experience, company-specific expectations, and other factors outside the proposed model.

Reliable evaluation also requires a sufficiently diverse dataset containing resumes, job descriptions, target roles, and manually verified skill relationships. Broader validation is therefore required before making claims about generalization across institutions or industries.

XX. Future Work

Several extensions can improve Career Compass AI. Large language models can be investigated for more detailed contextual extraction of skills and responsibilities from resumes and job descriptions. They may also provide explanations for why a particular skill has been prioritized.

A skill knowledge graph is another possible extension. Such a graph could represent relationships among roles, technologies, prerequisite skills, projects, and learning activities. This would allow the system to recommend a suitable sequence of skills rather than treating every missing competency independently.

The assessment module can also be made adaptive. When a student consistently performs well on beginner-level questions, the system can gradually introduce questions of greater difficulty. Practical coding exercises can provide another source of evidence beyond multiple-choice questions.

Future versions may also consider project portfolios, Git repositories, internship experience, interview preparation, and communication-related skills. At the institutional level, anonymized aggregate skill-gap information could help placement departments identify commonly missing competencies and design targeted training programmes.

XXI. Conclusion

Career Compass AI presents an integrated approach to student employability preparation by combining resume analysis, job-description matching, skill-gap identification, practical project recommendation, role-specific assessment, progress analytics, and personalized roadmap generation.

The central idea is to move beyond a one-time resume score. The student's current profile is first compared with a selected target role. The system then identifies important missing or weak competencies and connects them with practical development activities. After the student completes relevant preparation, the assessment module provides another source of information about current knowledge.

Retaining previous assessment attempts allows the system to observe changes rather than relying only on the latest result. These observations can be used to modify learning priorities and produce a more relevant roadmap for the next preparation stage.

The framework is therefore intended to support an iterative process of analysis, learning, assessment, and reassessment. The proposed methodology provides a foundation for a student-centred employability platform, while future experiments with diverse and manually validated datasets are required to quantify the effectiveness of each component.

XXII. *

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SUB-ID-779789

By Saravanan K

WORD COUNT 4692

TIME SUBMITTED 31-AUG-2026 01:45PM

PAPER ID 124037387

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Career Compass AI: An AI-Powered Employability Assessment and Career Readiness Platform Using Resume Analysis and Skill Gap Assessment

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***Abstract*—**
Students often learn several technologies during their studies, but they may not know how closely their current skills match the requirements of a particular job. A resume can contain programming languages, frameworks, projects, internships, and certifications, but these details alone do not show whether the student has enough practical knowledge. Career Compass AI is proposed as an integrated platform that connects resume analysis, job matching, skill-gap identification, project recommendation, technical assessment, progress tracking, and career-roadmap generation.

The system accepts a student's resume together with a job description or selected target role. Important information is extracted and organized before the profile is compared with the selected requirements. The analysis then identifies matched skills, missing skills, and areas that need improvement. Important skill gaps are connected with practical project suggestions so that students receive a clearer direction for developing the required competencies.

The platform also provides role-specific assessments to obtain additional evidence of technical understanding. Previous assessment attempts are retained so that changes in performance can be observed over time. Based on the latest results, learning priorities and the career roadmap can be updated. In this way, Career Compass AI treats employability preparation as a continuous process of analysis, learning, assessment, and reassessment rather than as a one-time resume evaluation.

***Index Terms*—Career readiness, resume analysis, skill gap analysis, job matching, natural language processing, technical assessment, project recommendation, career roadmap**

I. Introduction

During their studies, students tend to gain knowledge of several technologies without understanding the correlation of their skills to the job requirements. A resume may include details such as programming languages, frameworks, projects, internships, and certificates, but this information alone does not guarantee practical knowledge. Career Compass AI is a comprehensive system offering services such as resume analysis, job matching, skills gap identification, project suggestions, technical assessments, progress tracking, and career roads planning.

The system accepts a student resume and the job description of a target position. Then, relevant information is extracted from the resume and organized before the resume is compared with the selected job requirements. The analysis identifies the skills matching job requirements, lacking skills, and the skills requiring improvement. Along with necessary skills gap information, students get appropriate project recommendations.

The platform provides assessments based on the specific role to ensure more evidence of technical knowledge. The history of assessments is preserved so that progress can be monitored. Following new results, learning priorities and the career pathway can be changed. Thus, Career Compass AI treats employability preparation as a perpetual process of analysis and learning rather than a singular examination of one's resume.

Preparing for campus placements can be challenging because students need to do more than just create a resume. They have to understand the required skills for a job, pinpoint their shortcomings in their profile, and choose what should be learned prior to applying for the position. This process becomes even more complicated when students look for options among various job titles because different positions might require different sets of programming languages, frameworks, databases, cloud technologies, and methodologies.

A resume contains relevant information about education and qualifications in terms of skills, projects worked on, internships, and certificates. However, listing a technology does not mean having enough experience to work with it. The wording used in a job description may differ from similar requirements. For instance, one company may use the term "RESTful services" while another one will refer to "REST API". An algorithm based only on exact wording may miss the necessary correlations between different skills.

Techniques provided by Natural Language Processing play a significant role in the processing this form of unstructured information. The implementation of Transformer-based frame-

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works led to the introduction of contextual representations that could help resolve many issues facing people that require the processing of information in the field of natural language processing (NLP) [3]. BERT demonstrated the power of bidirectional into contextual representations for the improvement of understanding language [4]. Sentence-BERT made it possible to compare the meaning of sentences and fragments of texts in the least complicated manner, with the help of meaningful sentence embeddings [5]. The use of information retrieval technologies allows to represent documents and calculate their value [1].

The system links profile analysis with competence development. It matches the learner's profile with the selected goal position, pinpointing important qualifications gaps, proposing relevant projects to fill them and assessing the knowledge gained through specialized evaluations. Past results are saved for tracking progress and modifying future priorities.

II. Related Work

Automated recruitment as well as career services usually employ information retrieval, natural language processing, semantic similarity, machine learning as well as recommendation techniques. Information retrieval has provided the well-known techniques to describe documents and evaluate their relevancy to queries or documents related [1]. The learning to rank approaches also provide a good example of how to learn relevancy decisions.

It might also be true that transformers advanced the real capacity of NLP systems concerning the contextual relevance of words. BERT optimized this concept by allowing the use of bidirectional pre-training and getting impressive results in several cases of language comprehension. The Sentence-BERT method emerged later and allowed for comparing sentences using semantic embeddings. The rationale behind these methods is useful for employability application, mostly because the skills and responsibilities of a given position might be expressed by using different wording.

Random Forest has an ensemble learning method aimed at classifications and predictions [6]. XGBoost allows for creating an effective gradient boosting model for structured data [7]. This technique may be considered for willingness assessment when a labelled dataset is present.

Generally, existing methods relate only to different levels of career preparation process. For instance, a resume parsing software does not instruct the user on how to enhance their profile; a job matching system offers merely a score of similarity, while the way to proceed after that remains unclear; an evaluation tool assesses technical knowledge without linking it to the targeted job of a student.

In contrast, Career Compass AI integrates these steps into one workflow that allows conducting resume analysis, job matching, skill gap identification, project implementation, technical assessment, tracking progress, and adapting a roadmap.

III. Research Gap

The primary issue that this paper addresses is the gap in linking the resume-job match analysis with actual skill development.

Profile Analysis → Skill Diagnosis → Skill Development → Evaluation → Reevaluation

With this system the preparation stage of the student can be updated as new project or assessment evidence appears.

IV. Proposed System

Career Compass AI operates as a modular system with the result obtained from one step being used as an input for another step. The process starts from a student's CV and a job description or position of interest.

Currently available features are similar to a preparatory system rather than a one-time scoring system. Upon completion of the initial stage of analysis the produced outcomes are utilized to provide practical recommendations. Assessment results enter the process later on and are used to update the recommendations.

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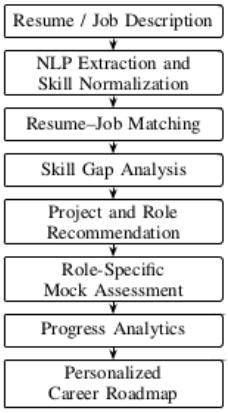


Fig. 1. Processing workflow of the proposed Career Compass AI framework.

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Where sufficient information is available, requirements can be separated into mandatory and preferred categories. This helps the system prioritize important requirements over optional ones.

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Similarly:

$$Amazon\ Web\ Services,\ AWS \rightarrow AWS \tag{2}$$

A normalization dictionary can map abbreviations, spelling variations, and common alternative expressions to a standard skill identifier.

D. Resume–Job Matching

Let S_r represent the normalized skills obtained from the resume and S_j represent the requirements obtained from the job description. A basic skill coverage value can be calculated as:

$$C = \frac{|S_r \cap S_j|}{|S_j|} \times 100 \tag{3}$$

where C represents the percentage of identified job requirements covered by the profile.

Keyword overlap may not be sufficient when different wording is used. Semantic representations can therefore be used in addition to direct matching. Cosine similarity between two embeddings is represented as:

$$Sim(E_r, E_j) = \frac{E_r \cdot E_j}{\|E_r\|\|E_j\|} \tag{4}$$

Here, E_r and E_j represent the corresponding vector representations.

VI. Skill Gap Analysis

After comparing the resume with the target role, the system identifies competencies that require attention. Instead of treating every unmatched term equally, the proposed approach divides the results into matched, missing, and improvement categories.

A matched skill is a competency represented in the student's profile. A missing skill is a requirement for which sufficient evidence is not found. An improvement skill is already present but may require stronger practical evidence or deeper understanding.

The priority of a gap depends on its importance to the selected role and the size of the observed gap. A conceptual priority score is:

$$P_s = R_s \times G_s \times I_s \tag{5}$$

where R_s represents role importance, G_s represents the estimated gap, and I_s represents relevance to the target position.

TABLE I
ILLUSTRATIVE SKILL GAP ANALYSIS

Skill	Status	Priority
Java	Match	–
Spring Boot	Match	–
MySQL	Match	–
Docker	Missing	High
AWS	Missing	High
Spring Security	Improve	Medium
Git	Improve	Medium

Note: The values shown are illustrative examples and are not experimental results.

VII. Project Recommendation

A conventional skill-gap report can identify what a student is missing but may not explain how that competency can be developed. Career Compass AI addresses this limitation by connecting identified gaps with practical project templates.

For example, when Docker, AWS, Spring Boot, and REST API development are identified as important gaps for a backend-oriented role, the platform can recommend an Employee Management System. The project can include a Spring Boot backend, MySQL database, REST endpoints, Docker containerization, Git-based version control, and cloud deployment.

The recommendation process can therefore be represented as:

Missing Skill → Project Template → Implementation Task → Skill Evidence

The purpose is not simply to increase the number of projects in a resume. The recommended project should provide an opportunity to apply the missing technology and create practical evidence of learning.

TABLE II
EXAMPLE PROJECT-TO-SKILL MAPPING

Project Component	Skill Developed
Backend Development	Spring Boot
Database Layer	MySQL
API Layer	REST API
Security	Spring Security
Containerization	Docker
Cloud Deployment	AWS
Version Control	Git

VIII. Suitable Job Role Recommendation

Students do not always have a single target position. A profile that is suitable for one role may require additional preparation for another. Career Compass AI can therefore compare the same profile with multiple predefined job roles.

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The role recommendation output is intended to indicate preparation alignment rather than predict whether a company will hire a student. The comparison considers matched skills, missing requirements, and, where available, assessment performance.

TABLE III
ILLUSTRATIVE JOB ROLE SUITABILITY

Job Role	Match	Level
Backend Developer	92%	Strong
Java Developer	88%	Strong
Full Stack Developer	81%	Good
Data Analyst	68%	Moderate
AI Engineer	60%	Improve

Note: These values are illustrative and should not be reported as measured experimental performance.

IX. Role-Specific Mock Assessment

A resume can indicate that a student has used a particular technology, but it cannot independently verify the level of understanding. The proposed assessment module provides an additional source of evidence through role-specific questions.

The assessment is selected according to the target role and important skills identified during the earlier analysis. For a Backend Developer profile, the question set may include Java, Spring Boot, SQL, REST APIs, Docker, Git, and cloud fundamentals.

The assessment process identifies the target role, obtains high-priority skills, selects suitable questions, conducts the test, calculates overall and skill-wise scores, and stores the attempt for later comparison.

This distinction between a skill being mentioned and a skill being demonstrated is an important part of the proposed framework.

TABLE IV
ILLUSTRATIVE SKILL-WISE ASSESSMENT HISTORY

Skill	Test 1	Test 2	Change
Java	65%	88%	+23
Spring Boot	40%	75%	+35
SQL	78%	84%	+6
Docker	20%	70%	+50
AWS	30%	65%	+35

X. Skill Analytics and Progress Tracking

Career preparation becomes more meaningful when changes can be observed over time. The proposed system stores individual assessment attempts instead of retaining only the latest score.

For a particular skill s, the change between two attempts can be represented as:

$$\Delta_s = S_{latest,s} - S_{previous,s} \tag{6}$$

where $S_{latest,s}$ is the latest score and $S_{previous,s}$ is an earlier score.

For example, an increase in the Docker assessment score from 20% to 70% corresponds to a 50 percentage-point improvement in assessment performance. This should not be interpreted as a direct measurement of real-world proficiency.

The analytics component can display the initial score, latest score, change over time, strongest skill, weakest skill, and remaining high-priority gaps. This gives students a simple way to decide whether their current preparation strategy needs to continue or change.

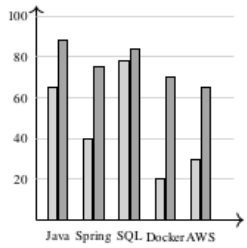


Fig. 2. Illustrative comparison of skill scores across assessment attempts.

XI. Personalized and Adaptive Career Roadmap

The roadmap is the stage where the analysis becomes a practical preparation plan. Instead of presenting students with a long list of unrelated learning resources, the system organizes activities around skills that are important for the selected role.

A typical roadmap begins with understanding the fundamentals of an identified skill, followed by suitable learning material and a practical task. The student can then apply the skill in a project, complete a role-specific assessment, review the result, and reassess the remaining gaps.

For example, if a student has reasonable Java knowledge but performs poorly in Docker and AWS, the next roadmap stage can focus on Docker fundamentals, containerizing a Spring Boot application, understanding basic AWS deployment concepts, and repeating the assessment.

The roadmap therefore changes according to newly observed evidence rather than remaining fixed after the initial resume analysis.

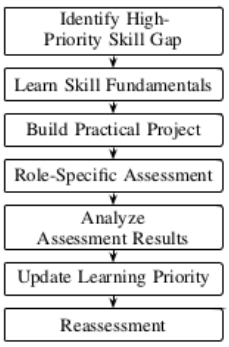


Fig. 3. Adaptive career-roadmap sequence driven by skill-gap and assessment feedback.

XII. Preparation Readiness Model

The system can provide a preparation-readiness indicator by combining observations from the student's profile, target role, and assessment performance. Unlike a simple resume

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similarity value, the indicator considers multiple sources of evidence.

A conceptual formulation is:

$$R = w_1 M + w_2 C + w_3 A \tag{7}$$

where M represents resume-job matching, C represents important skill coverage, A represents assessment performance, and w_1, w_2, w_3 are configurable weights.

The weights satisfy:

$$w_1 + w_2 + w_3 = 1 \tag{8}$$

The indicator is intended to describe preparation status rather than predict employment. Before being treated as a calibrated measure, the formulation should be tested with a suitable dataset.

Recruitment decisions depend on several factors outside the scope of this model, including interviews, coding performance, communication ability, work experience, and organization-specific selection procedures.

XIII. System Architecture and Implementation

The proposed platform follows a modular web architecture. The user interface provides facilities for resume upload, target-role selection, analysis results, assessment interaction, progress visualization, and roadmap presentation.

The application layer coordinates communication between the interface and processing modules. NLP processing handles resume and job-description text, while the matching engine compares normalized skills and semantic representations. The skill-gap module identifies missing and improvement areas. The recommendation engine connects these gaps with suitable projects, while the assessment engine manages role-specific tests.

A persistent database retains user profiles, extracted skills, target roles, assessment attempts, recommendations, and roadmap information. This structure allows previous results to be used when generating later recommendations.

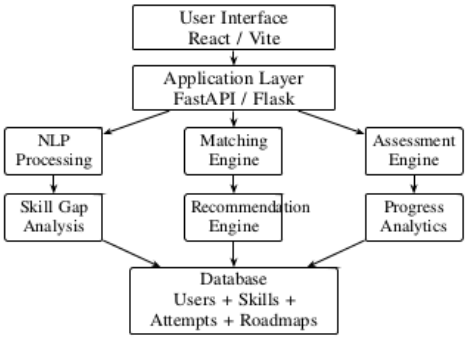


Fig. 4. Modular architecture of the proposed Career Compass AI platform.

A. Technology Stack

The implementation can use React with Vite for the web interface and a Python-based backend such as FastAPI or Flask for application services. Python libraries can support document processing, NLP operations, semantic comparison, and machine-learning components.

A relational database such as MySQL or PostgreSQL can store user profiles, extracted skills, job requirements, assessment attempts, project recommendations, and roadmap information. Chart.js or Recharts can be used to display skill-progress information through simple visualizations.

TABLE V
PROPOSED TECHNOLOGY STACK

Layer	Technology
Frontend	React.js, Vite
UI Styling	CSS / Tailwind CSS
Backend	FastAPI / Flask
NLP	Python, NLP Libraries
ML	Scikit-learn
Database	MySQL / PostgreSQL
Visualization	Chart.js / Recharts
Version Control	Git / GitHub

XIV. Data Model and Processing Flow

The database is designed to preserve information required for repeated analysis. A student may upload more than one resume and evaluate the profile against different target roles. Each target role can contain several required skills, and each assessment can generate multiple skill-wise results.

The principal entities include the user, resume, job profile, skill, skill gap, assessment, attempt, recommendation, and roadmap. The user entity stores basic account information, while the resume entity stores uploaded resume information and extracted content. The job profile contains the selected role and job-description information.

Skill-gap records maintain matched, missing, and improvement classifications. Assessment and attempt records preserve test information and individual results. Recommendation and roadmap records store suggested projects and learning priorities.

Maintaining these relationships allows the system to use earlier information when generating later recommendations. Assessment history, in particular, provides evidence for determining whether a skill area is improving or continues to require attention.

XV. Evaluation Methodology

Evaluation of Career Compass AI should be performed at both module and end-to-end levels. Since the platform performs several different tasks, a single accuracy value would not adequately describe its overall behaviour.

A. Resume Extraction Evaluation

Resume extraction can be evaluated using manually annotated resumes. Precision, recall, and F1-score can be calculated for information such as technical skills, education, experience, and certifications.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

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$$F1 = 2\frac{Precision \times Recall}{Precision + Recall} \quad (11)$$

These measures indicate how closely automatically extracted information agrees with manually verified information.

B. Skill Matching Evaluation

For skill matching, resume-job pairs can be manually annotated. Each requirement can be labelled according to whether the student's profile satisfies it. System predictions can then be compared with these reference labels.

C. Recommendation Evaluation

Project recommendations should be evaluated according to relevance. Evaluators can examine whether a recommended project addresses the identified skill gap and provides a realistic opportunity to apply the missing competency.

D. Assessment Evaluation

The assessment module can be evaluated in terms of question relevance, skill coverage, difficulty consistency, and repeated-attempt behaviour. Questions should remain aligned with the competencies expected for the selected role.

TABLE VI
EVALUATION PARAMETERS AND THEIR PURPOSE

Parameter	Evaluation Purpose
Resume Extraction	Correct extraction of profile entities
Skill Identification	Correct detection of relevant skills
Job Matching	Alignment with job requirements
Gap Detection	Identification of missing competencies
Recommendation	Relevance of suggested projects
Assessment	Reliability across attempts
Roadmap	Alignment with identified weaknesses
Progress Tracking	Measurement of score changes

XVI. Baseline and Ablation Study

An ablation study can be used to understand the contribution of the major modules. A simple baseline can begin with resume-job matching alone. Additional components can then be introduced progressively.

The first configuration performs profile matching only. The second adds skill-gap identification. The third includes role-specific assessment, while the final configuration adds the adaptive roadmap. The purpose is to determine whether the additional modules provide useful information beyond a conventional resume-matching approach.

TABLE VII
ABLATION CONFIGURATIONS

ID	Match	Gap	Test	Roadmap	Purpose
A	Yes	No	No	No	Baseline
B	Yes	Yes	No	No	Gap
C	Yes	Yes	Yes	No	Assessment
D	Yes	Yes	Yes	Yes	Complete

XVII. Results and Analysis Plan

The numerical performance of the proposed system should be reported only after testing with an appropriate dataset. The illustrative values shown in this paper demonstrate the intended presentation and should not be interpreted as measured experimental results.

The planned evaluation can include resume-extraction precision, recall and F1-score, skill-identification accuracy, agreement between automated job matching and human annotation, relevance of identified skill gaps, usefulness of project recommendations, assessment consistency, and changes in skill-wise scores across repeated attempts.

An important experiment is to compare the roadmap before and after a student completes an assessment. If the assessment shows improvement in one skill while another remains weak, the next roadmap should reflect that change. This provides a direct way to examine whether the adaptive component responds to newly available evidence.

XVIII. Discussion

The main strength of Career Compass AI is that it connects several stages of career preparation rather than treating resume analysis as the final result. A matching score can indicate how closely a profile resembles a job requirement, but it does not explain how the student can improve the profile.

The proposed workflow addresses this issue by converting identified gaps into practical development activities. When an important skill is missing, the system can prioritize it and connect it with a suitable project. The student can then use the project to gain implementation experience before taking another assessment.

The assessment component adds another useful distinction. A technology appearing in a resume is not necessarily equivalent to demonstrated knowledge. By evaluating role-specific questions and retaining previous attempts, the system obtains an additional signal that can be used when generating the roadmap.

The framework can also support multiple career directions. A student may have stronger preparation alignment with a Backend Developer role than an AI Engineer role. Instead of forcing a single recommendation, the system can show relative preparation levels and identify skills required to move toward another role.

The proposed framework should therefore be viewed as a career-preparation support system rather than an automated recruitment decision system.

XIX. Limitations

The proposed framework has several limitations. The first is the quality of input documents. Poorly formatted resumes, scanned pages, unusual layouts, or ambiguous terminology can reduce the accuracy of text and skill extraction.

A second limitation is that semantic similarity cannot prove practical proficiency. A student may mention a technology in a resume without having sufficient hands-on experience. The assessment module provides an additional signal, but even a structured test cannot completely replace practical project evaluation.

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Recommendation quality also depends on the underlying skill-to-project knowledge base. If a new technology or specialized role is not adequately represented, the platform may produce a less useful recommendation.

The preparation-readiness indicator should not be interpreted as a probability of employment. Actual recruitment depends on interviews, coding ability, communication, experience, company-specific expectations, and other factors outside the proposed model.

Reliable evaluation also requires a sufficiently diverse dataset containing resumes, job descriptions, target roles, and manually verified skill relationships. Broader validation is therefore required before making claims about generalization across institutions or industries.

XX. Future Work

Several extensions can improve Career Compass AI. Large language models can be investigated for more detailed contextual extraction of skills and responsibilities from resumes and job descriptions. They may also provide explanations for why a particular skill has been prioritized.

A skill knowledge graph is another possible extension. Such a graph could represent relationships among roles, technologies, prerequisite skills, projects, and learning activities. This would allow the system to recommend a suitable sequence of skills rather than treating every missing competency independently.

The assessment module can also be made adaptive. When a student consistently performs well on beginner-level questions, the system can gradually introduce questions of greater difficulty. Practical coding exercises can provide another source of evidence beyond multiple-choice questions.

Future versions may also consider project portfolios, Git repositories, internship experience, interview preparation, and communication-related skills. At the institutional level, anonymized aggregate skill-gap information could help placement departments identify commonly missing competencies and design targeted training programmes.

XXI. Conclusion

Career Compass AI presents an integrated approach to student employability preparation by combining resume analysis, job-description matching, skill-gap identification, practical project recommendation, role-specific assessment, progress analytics, and personalized roadmap generation.

The central idea is to move beyond a one-time resume score. The student's current profile is first compared with a selected target role. The system then identifies important missing or weak competencies and connects them with practical development activities. After the student completes relevant preparation, the assessment module provides another source of information about current knowledge.

Retaining previous assessment attempts allows the system to observe changes rather than relying only on the latest result. These observations can be used to modify learning priorities and produce a more relevant roadmap for the next preparation stage.

The framework is therefore intended to support an iterative process of analysis, learning, assessment, and reassessment. The proposed methodology provides a foundation for a student-centred employability platform, while future experiments with diverse and manually validated datasets are required to quantify the effectiveness of each component.

XXII. *

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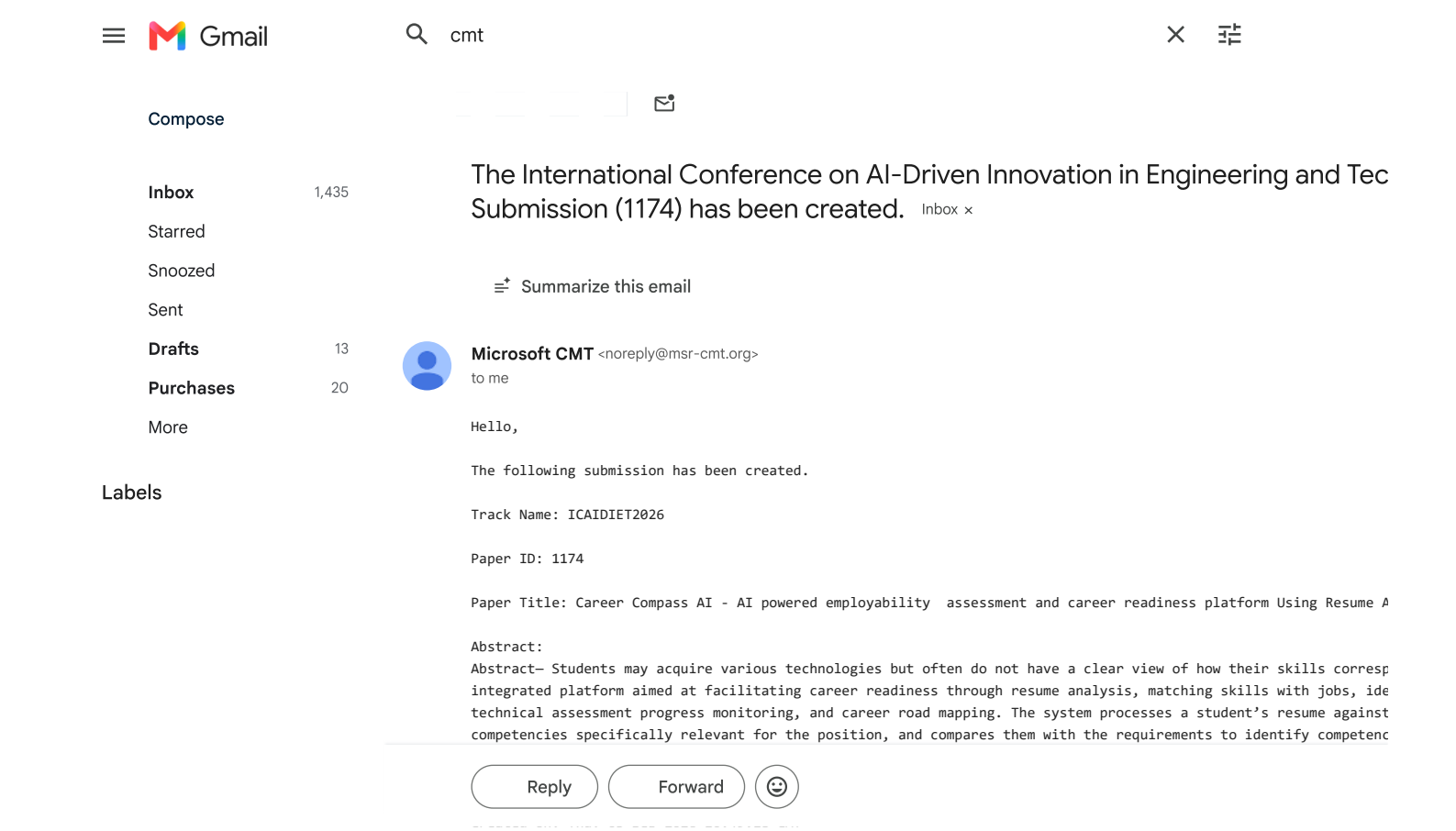
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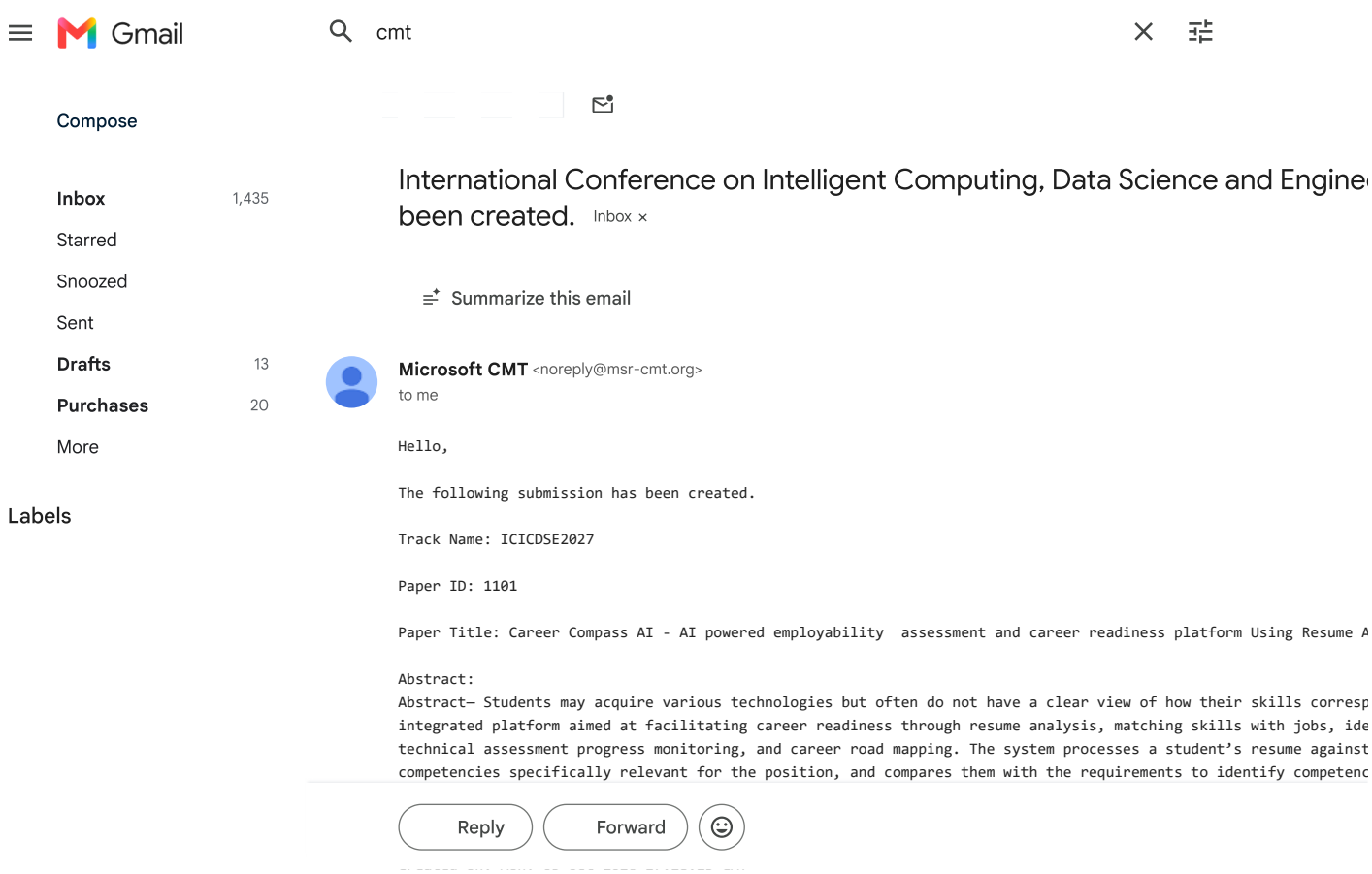
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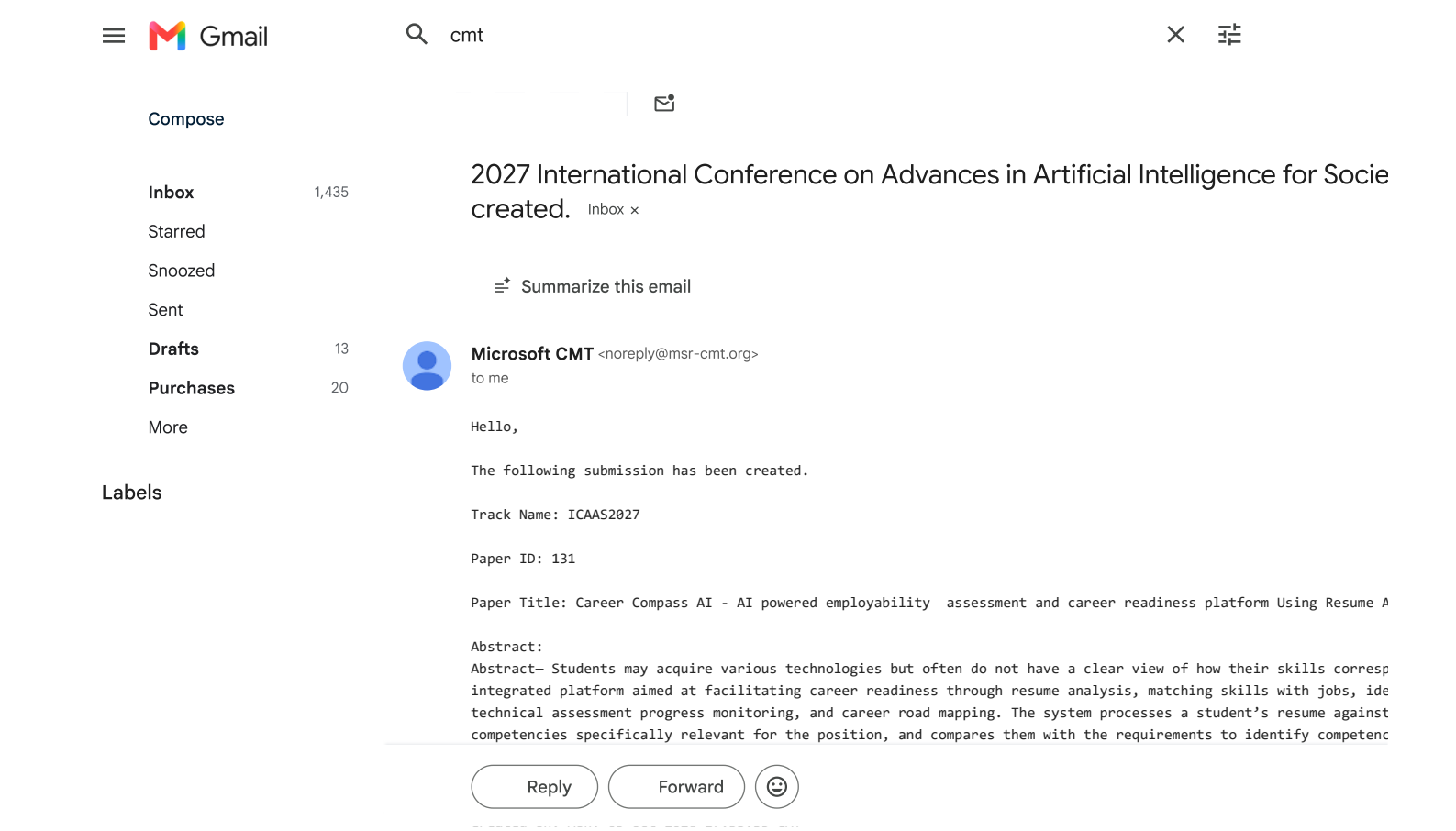
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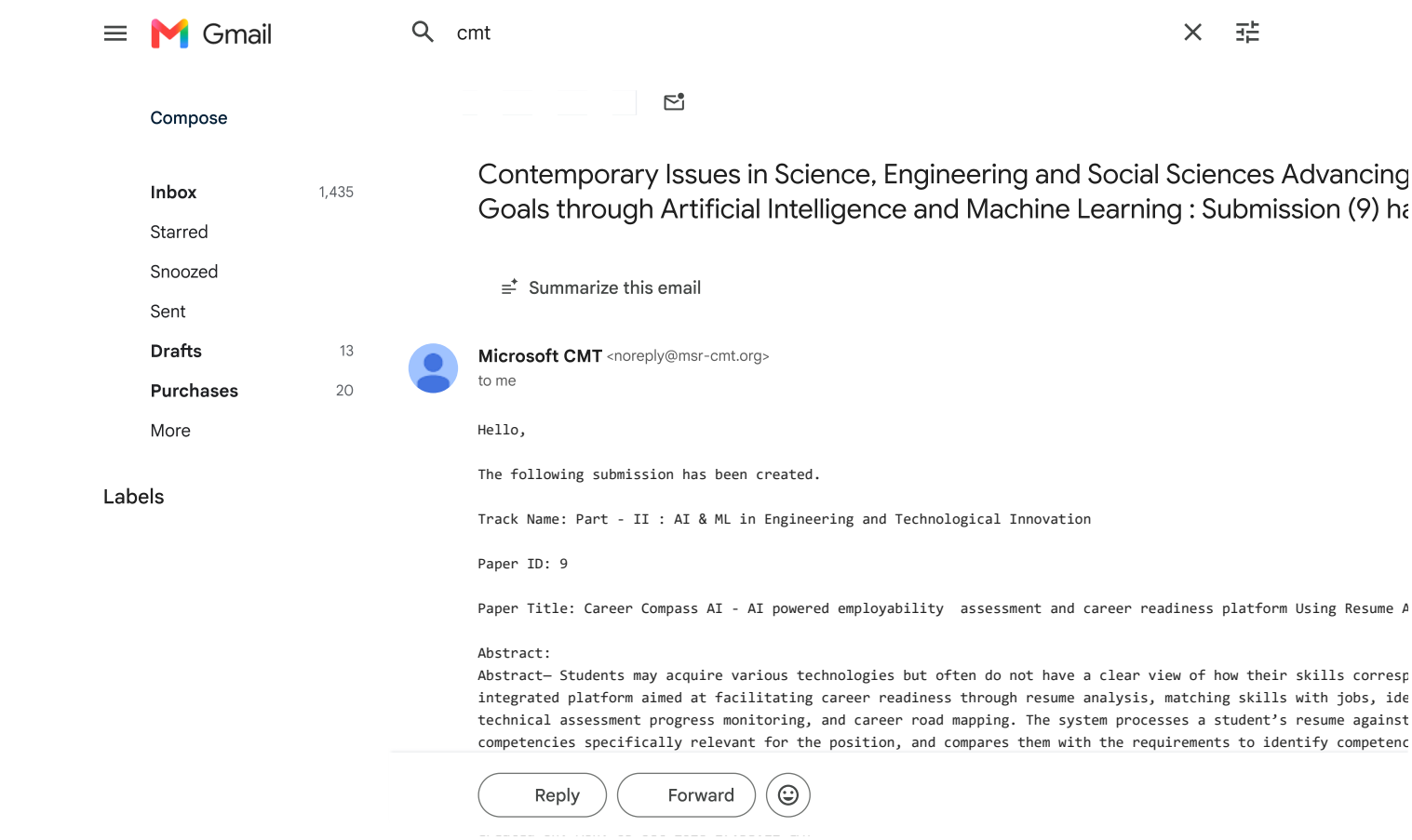
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